

IN-SPACe CANSAT India Student Competition 2024-25 (2nd Edition)

Preliminary Design Review (PDR)

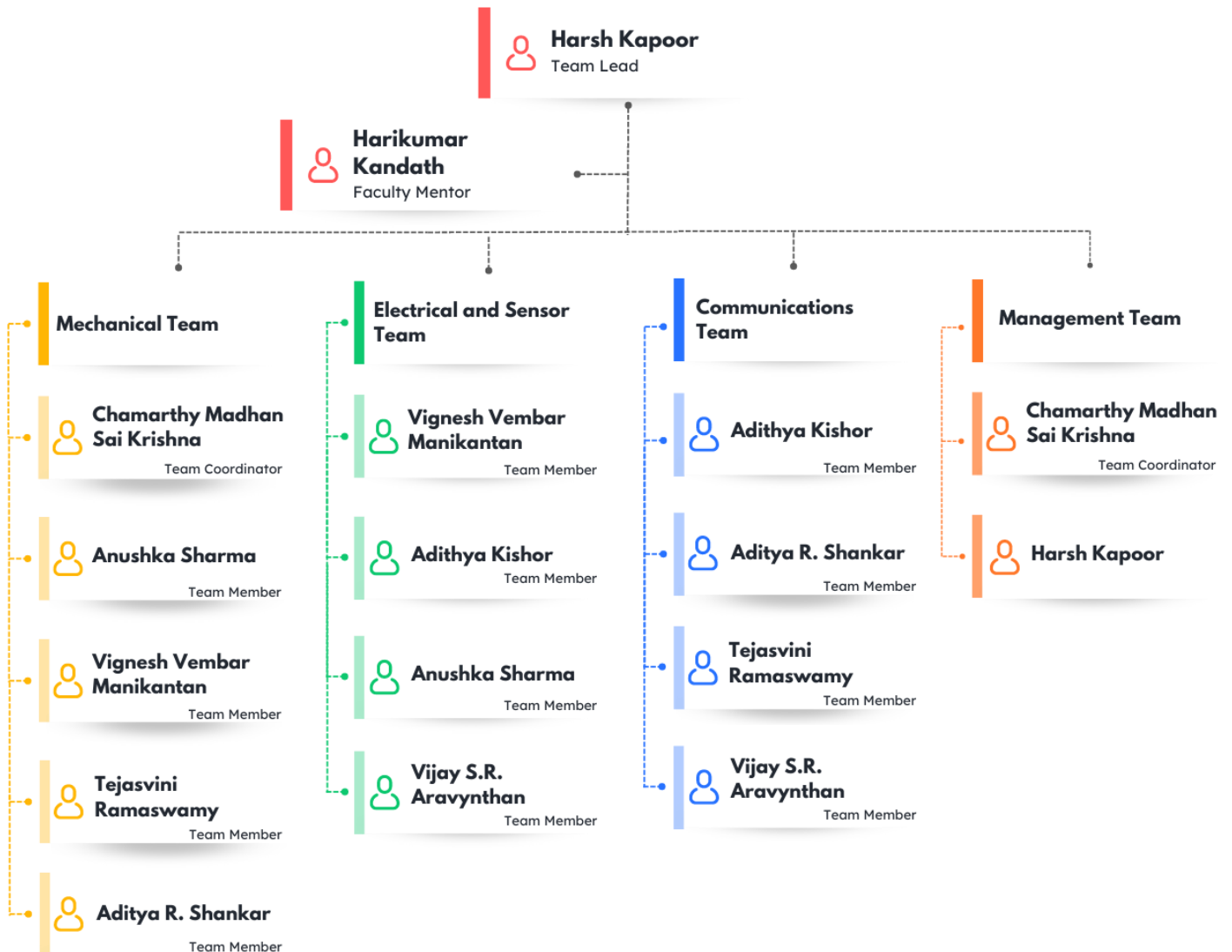
2024-ASI-CANSAT-044-BRAHMAND

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Team Organization

TEAM ORGANIZATION



Team Member Names	Roles	Responsibilities
Harsh Kapoor	Team Lead, Management Team Member	Communications with college, faculty, organizers, and team.
Madhan Sai Krishna	Team Coordinator, Management and Mechanical Team Member	Inter-Team Coordination, 3D modelling and designing.
Anushka Sharma	Mechanical, Sensor and EPS Team Member	Selection of Mechanisms and Sensors for the mission.
Adithya Kishor	Communications, Sensor and EPS Team Member	Flight Software and Communications management.
Aditya R Shankar	Mechanical and Communications Team Member	Gyroscope and Communications management.
Tejasvini Ramaswamy	Mechanical and Communications Team Member	Materials selection, 3D modelling and Communication management.
Vignesh Vembar Manikantan	Mechanical, EPS and Sensor Team Member	Gyroscope and Sensors selections.
Vijay Aravyanthan SR	Communications, Sensor and EPS Team Member	Sensors and Communication System management.

In addition to this, we are pleased to announce that **Dr. Harikumar Kandath** has graciously accepted the role of Faculty Advisor for our team.

System Overview

Mission Objectives

Main Objectives	
<i>Design an “Innovative Mechanical Gyro-control” system that shall demonstrate the descent control of the CANSAT</i>	
1.	Weight and Cost Design: The CanSat shall have a mass of 1000 +/- 50 grams, meet specific dimensional requirements and survive structural and environmental forces. Ensure the cost remains under ₹1,00,000.
2.	Dimensions: CanSat shall fit in a cylinder of 0.15m diameter and 0.40m height and have tolerances to ensure deployment from rocket fairing.
3.	Gyroscope: CanSat shall stabilize itself during the decent using the Mechanical Gyroscopes.
4.	Structural Integrity: CanSat structure shall be built to survive 15 Gs of launch acceleration & 30 Gs of shock. All electronics shall be enclosed and shielded from the environment. No electronics can be exposed except for sensors. There must be a structural enclosure.
5.	Electronics and Telemetry: The CanSat shall continuously sense and transmit a variety of sensory data from launch and stop this transmission upon landing.
6.	Ground Station and Flight Software: Develop a ground station that calibrates altitude, generates CSV files, and displays telemetry in real-time during descent. The software must also implement a "Simulation mode".

Bonus Objectives

1.	Novel descent trajectory Control, Innovative materials.
2.	Additional Innovative Sensors and Communication Systems.
3.	Provision of video capture from separation till final touchdown.
4.	Innovative recovery techniques viz. HAM radio/Advanced beacons.
5.	Innovative Quality and reliability analysis methodologies.

Rationale for attempting Bonus Objectives

1. **IIIT Hyderabad** being a research focused institution has expertise in various fields (innovative materials and sensors) and we at Team Brahmand believes that we can leverage this research into real world applications.
2. A video camera can be also easily implemented given our choice of battery and flight controller and a provision to store the video on an **onboard SD Card** can be made. Provision to have a live video feed needs additional testing and planning.

Mission Requirement Summary

Every mission requirement has been mapped to a Subsystem Level Team and Assigned an ID for better tracking of the requirements. A (Analysis), I (Inspection), T (Testing), and D (Demonstration) verification method has been used to check the progress of every requirement in the physical CANSAT. The table of Requirements is as follows –

Req No.	ID	Description	Subsystem	Verification Method			
				A	I	T	D
1	BR1	Total mass of the CANSAT shall be under 1.0 Kg (+/- 0.050 Kg).	Mechanical	✓			
2	BR2	CANSAT shall fit in a cylindrical body of 0.15 m diameter x 0.40 m height. Tolerances are to be included to facilitate container deployment from the rocket fairing.	Mechanical	✓			
3	BR3	Teams are free to use any design process and material for the cylinder development.	Mechanical	✓	✓		
4	BR4	Any sharp edges on the container body shall be avoided as it can cause problems during the CANSAT ejection from the rocket.	Mechanical	✓			
5	BR5	For ease in location of the CANSAT, the color of the CANSAT body shall be painted with fluorescent colors i.e., pink, red or orange, and shall embody the Indian flag.	Mechanical	✓	✓		
6	BR6	Rocket Airframe will not be allowed to be used as a part of any CANSAT operation.	Mechanical	✓			
7	BR7	Additional weightage will be given to the MADE IN INDIA components in the making of CANSAT.	All	✓			
8	BR8	The Global Navigation Satellite System (GNSS) sensor that includes Navigation by Indian Constellation (NavIC) will be given extra weightage.	Communications	✓	✓		

9	BR9	The CANSAT shall consist of necessary sensors to provide the following mandatory Real-time datasets: Position data, altitude, pressure, temperature, orientation data, power data & system status.	Electronics	✓	✓		
10	BR10	The data shall be displayed in near real-time on the ground station User Interface/Software. Teams are free to use any design to build their own user interface.	Communications	✓			
11	BR11	CANSAT software shall have provision to record the data and save it into an onboard SD card, in case of telemetry connection loss during flight.	Communications	✓	✓		
12	SR1	All electronics shall be enclosed and shielded from the environment. No electronics can be exposed except for sensors. There must be a structural enclosure.	Mechanical	✓			
13	SR2	CANSAT structure shall be built to survive 15 Gs of launch acceleration & 30 Gs of shock.	Mechanical	✓			
14	SR3	Electronic circuit boards must be hard mounted using proper mounts such as standoffs and screws. High-performance adhesives can also be used.	Mechanical	✓			
15	SR4	Team number, email address and phone number must be placed on the structure in English, Hindi and the regional language of the launch state to aid in recovery.	Mechanical	✓			
16	SR5	An audio beacon shall be installed on CANSAT as a recovery assist. It may be powered after landing or operate continuously. The audio beacon must have a minimum sound pressure level of 92 dB, unobstructed	Electrical & Power	✓			

17	PR1	The CANSAT shall have an external power switch with an indicator light or sound for being turned on or off, to avoid the disassembling of CANSATs on the launch pad.	Electrical & Power Mechanical	✓			
18	PR2	The CANSAT shall have a battery capacity to support up to 2 hours of wait in on the launch pad with additional time for flight operations.	Electrical & Power Electronics	✓			
19	PR3	The battery source may be alkaline, Ni-Cad, Ni-MH or Lithium ion. Lithium polymer batteries are not allowed. Lithium cells must be manufactured with a metal package similar to 18650 cells.	Electrical & Power Electronics	✓			
20	PR4	An easily accessible battery compartment must be included allowing batteries to be installed or removed in less than a minute and not require total disassembly of the CANSAT.	Electrical & Power Mechanical	✓			
21	PR5	Spring contacts shall not be used for making electrical connections to batteries. Care must be taken as the shock forces can cause momentary disconnects of power.	Electrical & Power Mechanical	✓			
22	DDR1	The CANSAT shall contain a total of 2 descent control mechanisms, to be used at different stages while descent.	Mechanical	✓			
23	DDR2	1st mechanism shall be a parachute while the second mechanism shall be decided by the teams. Few examples for the second descent control mechanism are: secondary parachute/streamer/glider/rotor mechanism etc.	Mechanical	✓			
24	DDR3	CANSAT shall immediately deploy the 1st parachute after ejection from the rocket.	Mechanical	✓			

25	DRR4	The 1st parachute shall be connected to the outer body of the CANSAT and no ejection mechanism shall be attached to it.	Mechanical	✓			
26	DRR5	The descent rate of the 1st parachute shall be 20 m/s (+/- 5m/s).	Mechanical	✓			
27	DRR6	The second descent control mechanism shall open at an altitude of 500 m (+/-10 m) to further decrease the descent rate of the CANSAT to 1 to 3m/s	Mechanical	✓			
28	DRR7	The descent control system shall not use any hazardous chemical-based explosive or pyrotechnic devices. However, green propulsion is allowed if being used under the same weight constraint.	Mechanical	✓			
29	DRR8	CANSAT shall stabilize itself during the decent using the Mechanical Gyroscopes.	Mechanical	✓			
30	DRR9	The descent control mechanism & all the attached components in CANSAT shall survive 30 Gs of shock at the time of launch and separation.	Mechanical	✓			
31	DRR10	All the electronic components shall be enclosed and shielded from the environment except for sensors.	Mechanical	✓			
32	CR1	The CANSAT communications radio shall be the XBEE radio series 1/2/pro.	Communications	✓			
33	CR2	The XBEE radios shall have their NETID/PANID set to the team number.	Communications	✓			
34	CR3	The XBEE radio shall not use the broadcast mode	Communications	✓			
35	CR4	The XBEE radio can operate in any mode as long as it does not interfere with other XBEE radios.	Communications	✓			
36	GS1	Each team shall develop and use their own ground station. All	Communications	✓			

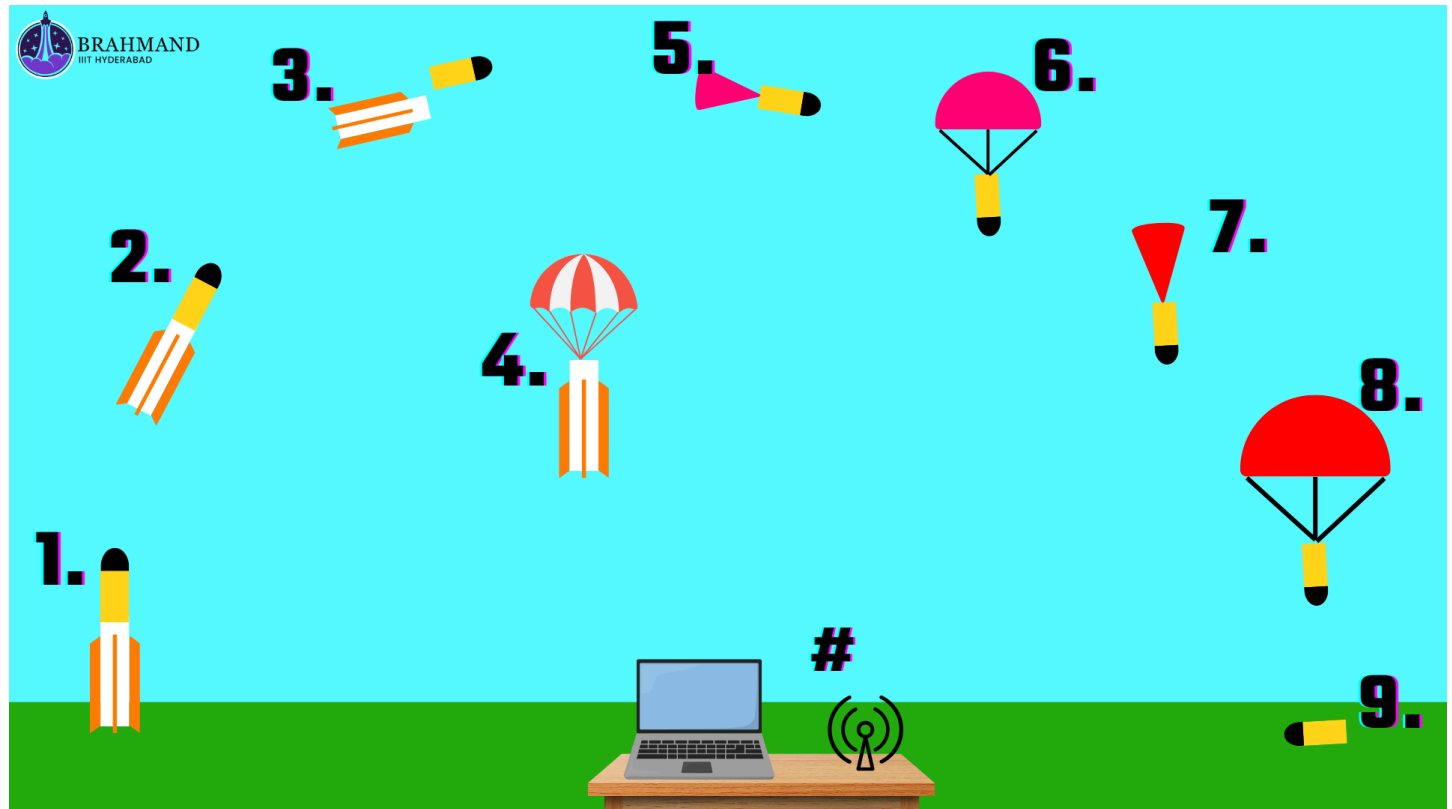
		telemetry shall be displayed in near real-time during launch and descent. All telemetry shall be displayed in international system of units i.e., SI system. Teams shall plot data in real-time during flight.					
37	GS2	The ground station shall command the CANSAT to start transmitting telemetry prior to launch.	Communications	✓			
38	GS3	The ground control station antenna shall be elevated from ground level to ensure adequate coverage and range.	Communications	✓			
39	GS4	Stability of the ground station must be ensured.	Communications	✓			
40	GS5	The CANSAT shall not transmit telemetry until commanded by the team. Command can be executed while the CANSAT is in the rocket on the launch pad.	Communications	✓			
41	GS6	The ground station shall be able to command the CANSAT to calibrate gyros, barometric altitude, accelerometer to command the parameters to zero as the CANSAT sits on the launch pad.	Communications	✓			
42	GS7	The ground station shall generate .csv files of all sensor data as specified in the Telemetry Requirements section.	Communications	✓			
43	GS8	Telemetry shall include mission time with one second or better resolution.	Communications	✓	✓		
44	GS9	In case of processor reset the mission clock/time stamp & system state shall be maintained.	Communications	✓			
45	GS10	Teams shall plot each telemetry data field in real-time during flight.	Communications	✓			
46	GS11	The ground station shall include one laptop computer with a minimum of two hours of battery	Communications	✓			

		operation, XBEE radio and a hand-held antenna.					
47	GS12	The ground station must be portable so the team can be positioned at the ground station operation site along the flight line and if required the team can also move to a different location, in case of distant landing location, in order to locate the CANSAT.	Communications	✓			
48	FS1	The flight software shall maintain and telemeter an indicator of the CANSAT flight software state.	Communications	✓			
49	FS2	In the event of a processor reset during the mission, the flight software shall be able to determine the correct state.	Communications	✓			
50	FS3	The states shall be described in the review presentation by each team.	Communications	✓			
51	T1	Upon powering up, the CANSAT shall collect the required telemetry at a 1 Hz sample rate or better. The telemetry data shall be transmitted with ASCII comma separated fields followed by a carriage The flight software shall maintain and telemeter an indicator of the CANSAT flight software state. An example set of states is 0 (BOOT), 1 (TEST_MODE), 2 (LAUNCH_PAD), 3 (ASCENT), 4 (ROCKET_DEPLOY), 5 (DESCENT), 6 (AEROBREAK_RELEASE), and 7 (IMPACT).	Communications	✓			
52	T2	The received telemetry for the entire mission shall be saved on the ground station computer as a comma-separated value (.csv) file that will be examined by the competition judges. Teams will provide the file to the judges	Communications	✓			

		immediately after the launch operations via USB drive. The .csv file shall include headers specifying each field of data.					
53	T3	The telemetry parameters display format with resolution needs to be provided. The telemetry parameters display format with resolution needs to be provided as given in the table in the Mission Requirements Document of PDR.	Communications	✓	✓		

System Concept of Operations (CONOPS)

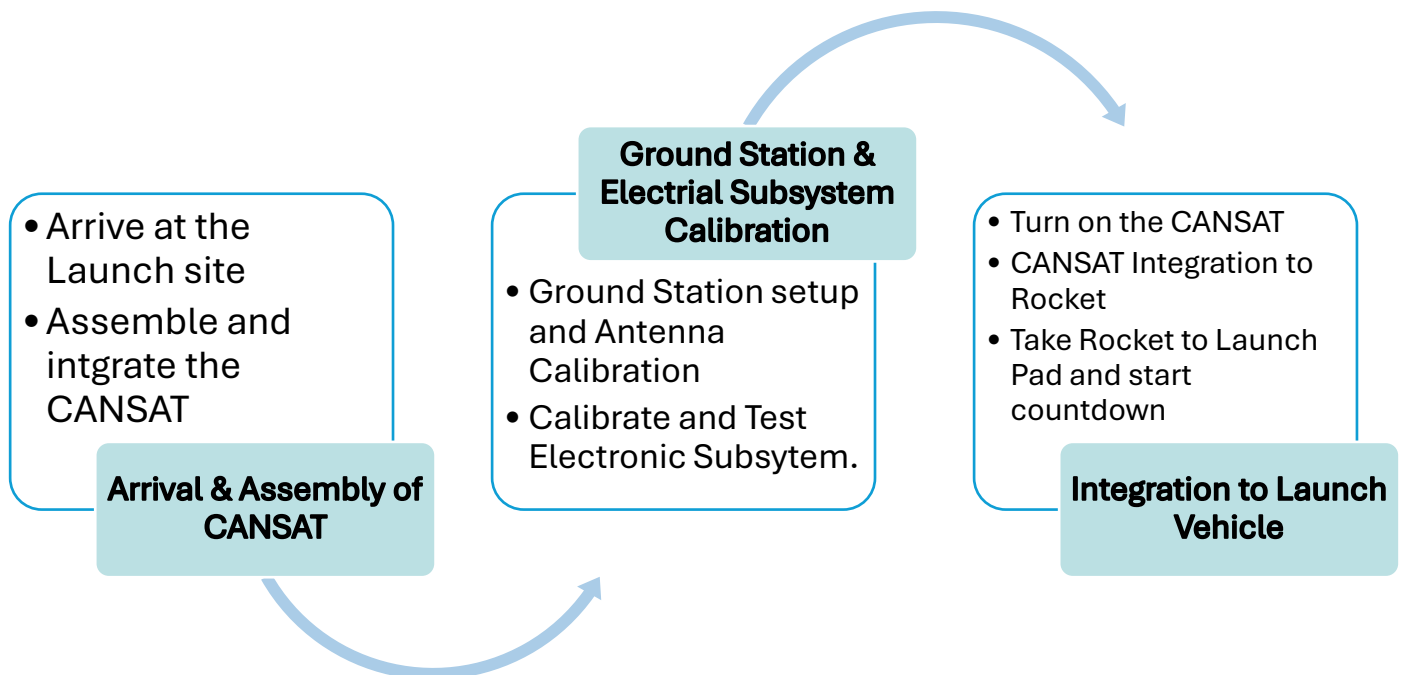
Mission CONOPS –



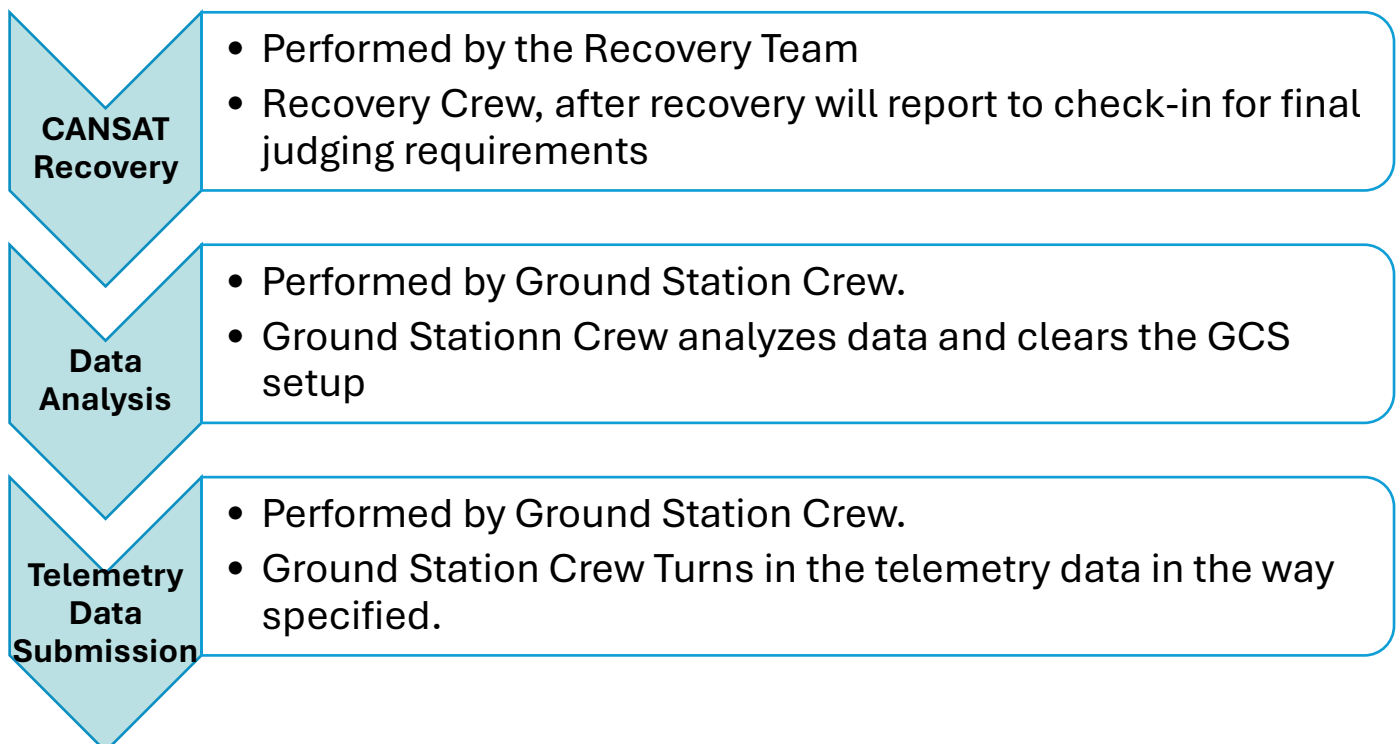
<u>Sr. no.</u>	<u>Description</u>
1.	Launch (0m)
2.	In Flight (0m – 1000m)
3.	Separation from rocket at apogee and stability mechanism starts (1000m)
4.	Rocket Descends
5.	Deployment of first parachute (1000m)
6.	Descent with first parachute (1000m – 500m)
7.	Deployment of secondary descent control mechanism (500m)
8.	Descent with secondary mechanism (500m – 0m)
9.	Landing and Retrieval (0m)
#	Ground Station and Antenna

This is a the currently planned mission based on the current mission description. We will have dedicated Assembly, Ground Station and Recovery teams on the day of launch that will handle their respective jobs for a smooth launch day.

Pre-Launch Activities –



Post-Launch Activities –

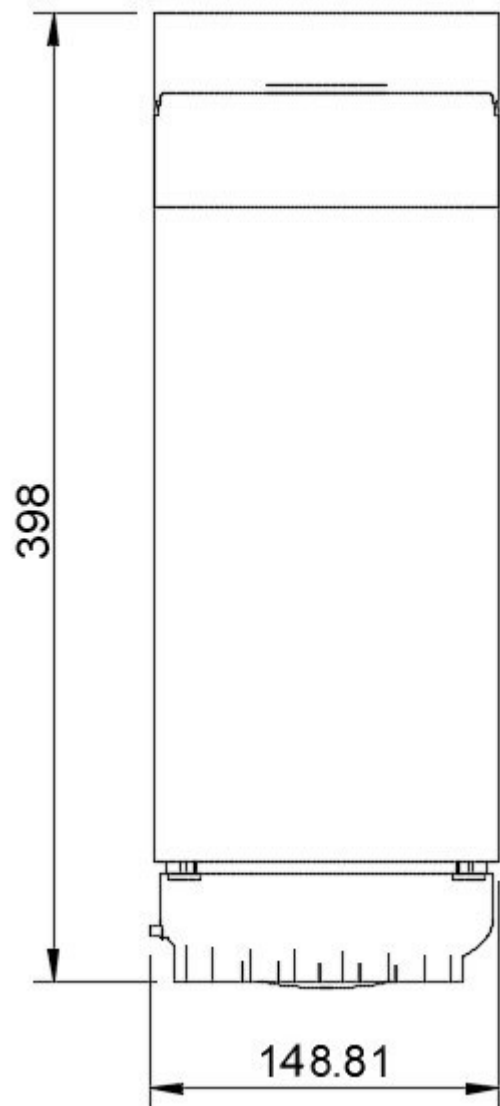
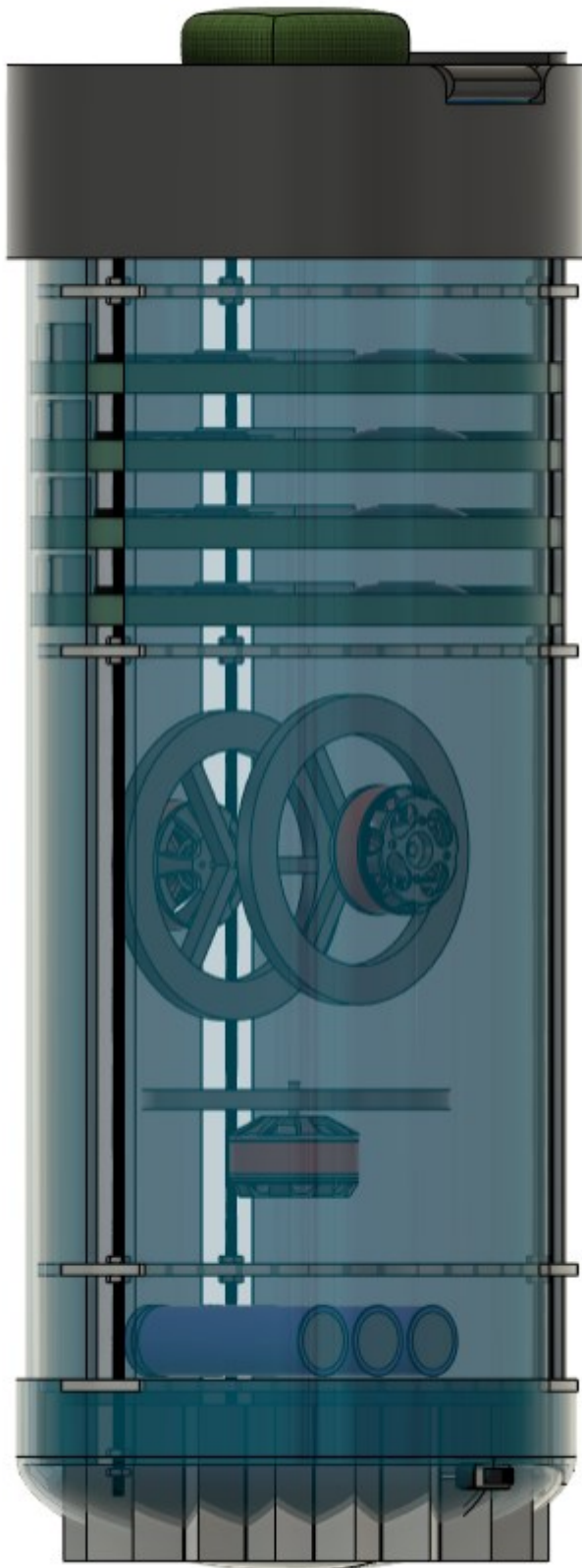


Colour Coding Scheme of CANSAT Components –

Sr No	Parts/Location	Colour			Reasoning
1.	Outer Casing	Saffron	White	Green	Enhanced Visibility and Ease of tracking during Descent and Recovery
2.	Electronics and Electricals	Fluorescent Green			Ease of Locating by Visual Inspection
3.	First Parachute	Fluorescent Pink			Mission Requirement and Ease of Tracking
4.	Second Parachute	Fluorescent Red			Mission Requirement and Ease of Tracking
5.	Gyroscope and Other Mechanical Components	Grey	White		Ease of Locating by Visual Inspection
6.	Lower Shell	Transparent			Required for Video Recording
7.	Structural Elements	Black			Ease of Locating by Visual Inspection

The colour coding is made keeping in mind the mission requirements and for ease in identifying the components of various teams, and ease in tracking of the CANSAT during descent and recovery.

CANSAT Physical Layout



System Requirement Compliance

Re q. No .	Requirements	Compliance
1.	Total mass of the CANSAT shall be under 1000gm (+/- 50 gm).	Comply
2.	CANSAT shall fit in a cylindrical body of 0.15 m diameter x 0.40 m height. Tolerances are to be included to facilitate container deployment from the rocket fairing.	Comply
3.	Any sharp edges on the container body shall be avoided as it can cause interfere during the CANSAT ejection from the rocket.	Comply
4.	Colour of the CANSAT body shall be fluorescent i.e., pink, red or orange, and shall embody the Indian flag.	Comply
5.	Rocket Airframe will not be allowed to be used as a part of any CANSAT operation.	Comply
6.	The CANSAT shall consist of necessary sensors to provide the following mandatory Real-time datasets: Position data, altitude, pressure, temperature, orientation data, power data & system status.	Comply
7.	Each data field shall be displayed in real-time on the ground station user interface/software.	Comply
8.	CANSAT shall also record the data and save it into an onboard SD card in case of telemetry connection loss	Comply
9.	All electronics shall be enclosed and shielded from the environment. No electronics can be exposed except for sensors. There must be a structural enclosure.	Comply

10 .	CANSAT structure shall be built to survive 15 Gs of launch acceleration & 30 Gs of shock.	Comply
11 .	Electronic circuit boards must be hard mounted using proper mounts such as standoffs and screws. High-performance adhesives can also be used.	Comply
12 .	Team number, email address and phone number must be placed on the structure in English, Hindi and the Regional language of the launch state to aid in recovery.	Comply
13 .	An audio beacon shall be installed on CANSAT as a recovery assist. It may be powered after landing or operate continuously. The audio beacon must have a minimum sound pressure level of 92 dB, unobstructed.	Comply
14 .	The CANSAT shall have an external power switch with an indicator light or sound for being turned on or off, in order to avoid the de-assembling of CANSATs on the launch pad.	Comply
15 .	The CANSAT shall have a battery capacity to support up to 2 hours of wait in on the launch pad with additional time for flight operations.	Comply
16 .	The battery source may be alkaline, Ni-Cad, Ni-MH or Lithium ion. Lithium polymer batteries are not allowed. Lithium cells must be manufactured with a metal package similar to 18650 cells.	Comply
17 .	An easily accessible battery compartment must be included allowing batteries to be installed or removed in less than a minute and not require total disassembly of the CANSAT.	Comply
18 .	Spring contacts shall not be used for making electrical connections to	Comply

	batteries. Shock forces can cause momentary disconnects.	
19	The CANSAT shall contain a total of 2 descent control mechanisms, to be used at different stages while descent.	Comply
20	CANSAT shall immediately deploy the first parachute after ejection from the rocket.	Comply
21	The first parachute shall be connected to the outer body of the CANSAT and no ejection mechanism shall be attached to it.	Comply
22	The descent rate of the 1st parachute shall be 20 m/s +/- 5m/s	Comply
23	The second descent control mechanism shall open at an altitude of 500m (+/-10 m) to further decrease the descent rate of the CANSAT to 1 to 3m/s	Comply
24	The descent control system shall not use any hazardous chemical-based explosive or pyrotechnic devices. However, green propulsion is allowed if being used under the same weight constraint.	Comply
25	CANSAT shall stabilize itself during the decent using the mechanical gyro mechanism.	Comply
26	The CANSAT communications radio shall be the XBEE radio series 1/2/pro.	Comply
27	The XBEE radios shall have their NETID/PANID set to the team number.	Comply
28	The XBEE radio shall not use the broadcast mode.	Comply
29	The XBEE radio can operate in any mode as long as it does not interfere with other XBEE radios.	Comply
30	Each team shall develop and use their own ground station. All telemetry shall be displayed in real-time during launch and descent. All telemetry shall be displayed in engineering units (meters,	Comply

	meters per second, Celsius, etc.). Teams shall plot data in real-time during flight.	
31	The ground station shall command the CANSAT to start transmitting telemetry prior to launch.	Comply
32	The ground control station antenna shall be elevated from ground level to ensure adequate coverage and range.	Comply
33	Stability of the ground station must be ensured.	Comply
34	The CANSAT shall not transmit telemetry until commanded by the team ground station. Command can be executed while the CANSAT is in the rocket on the launch pad.	Comply
35	The ground station shall be able to command the CANSAT to calibrate gyros, barometric altitude, accelerometer to command the parameters to zero as the CANSAT sits on the launch pad.	Comply
36	The ground station shall generate .csv files of all sensor data as specified in the Telemetry Requirements section.	Comply
37	Telemetry shall include mission time with one second or better resolution.	Comply
38	Mission time/timestamp and system status states shall not be affected in the event of a processor reset during the launch and mission.	Comply
39	The ground station shall include one laptop computer with a minimum of two hours of battery operation, XBEE radio and a hand-held antenna.	Comply
40	The ground station must be portable so the team can be positioned at the ground station operation site along the flight line and if required the team can also move to a different location in case of distant landing location in order to locate the CANSAT.	Comply

41 .	The flight software shall maintain and telemeter an indicator of the CANSAT flight software state. An example set of states is 0 (BOOT), 1 (TEST_MODE), 2 (LAUNCH_PAD), 3 (ASCENT), 4 (ROCKET_DEPLOY), 5 (DESCENT), 6 (AEROBREAK_RELEASE), and 7 (IMPACT).	Comply
42 .	Upon powering up, the CANSAT shall collect the required telemetry at a 1 Hz sample rate or more. The telemetry data shall be transmitted with ASCII comma-separated fields followed by a carriage return	Comply

Sensor Subsystem Overview

Sensor Subsystem Overview

Outline

We have chosen sensors so that one sensor is allocated per measurement and having some sensor modules which measure multiple fields to add redundancy in the measurements taken. We have divided them into two parts – Core and Optional sensors. Core sensors include all the sensors which measure the fields listed in the Mission Requirement Document while the Optional Sensors include the sensors, we plan to incorporate in the CanSat for additional measurements.

Core

- GPS
- Barometric Sensor
- Temperature Sensor
- Humidity Sensor
- Voltage Monitoring Sensor
- 6/9 axis IMU

Optional

- Multi Module Sensors
- High G Accelerometer
- Camera
- Air Quality Sensors
- Air Speed Sensors

The following criteria were used for selecting the sensors -

- Low power consumption
- Small size
- Availability
- Documentation
- Shock and thermally tolerant
- Fast start times

GPS Sensor

The following are the requirements from the GPS Sensor -

- Good accuracy (location and speed)
- Good altitude to be cross-referenced with barometer
- NAVIC compatibility
- Ease of use with
- Reliability in high vibration environment
- Low LOS (Loss of signal) and multipath for accurate reading at low altitudes.
- Fast hot and warm start [Refer appendix [A1](#)]

Different GPS Sensors considered

Name	Bands	Positional Accuracy (m, CEP)	Velocity Accuracy (cm/s)	Update rate (Hz)	Hot start (Sec)	Size (cm x cm)	Current (mA)	Inter-face	Cost (INR)
Neo M9	GPS GLONASS Galileo BeiDou	2-4	5	25	2	1.2 x 1.6	50	UBX, I2C, UART	3249
LC89 HA	GPS, GLONASS , Galileo, BeiDou NaViC	<1.8	<0.1	10	<1	1.4 x 1.6	99	UART, I2C, SPI	900
ELNMC1A	GPS, GLONASS , Galileo, BeiDou NaViC	1.5	10	1	1-2	2.2 x 2.4	50	UART, I2C, SPI	550
BD9250s	GPS, GLONASS , Galileo, BeiDou, NaViC	<1 m	-	20	< 2	2 x 2.5	425 (1.4 W)	UART	10000
Venus827	GPS, NaViC	2.5 (-156 dBm)	0.1	20	1	1.5 x 1.5	50	UART LVTTTL	2000

* All modules need 3.3 V to operate

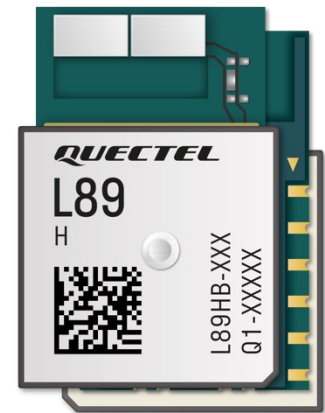
* CEP -> Circular Error Probability or CEP is defined as **the radius of a circle centered on the true value that contains 50% of the actual GPS measurements** [[Appendix 1.B](#)]

Selected GPS - **Quectel LC89HA**

Multi-GNSS engines for GPS, **IRNSS (NaViC)**, GLONASS, BeiDou, Galileo and QZSS

Reasons:

- NaViC support
- Good documentation
- Compatibility with ESP32
- Availability of breakout boards [\[1\]](#)
- Low power consumption
- Multipath + external antenna support



Interface

- I2C Interface: Max. bit rate up to 400 kbps
- **UART** Interface: Default: 9600 bps
- Update Rate: 1 Hz (Default), up to 10 Hz*



Power and Weight Requirements

- 3.1 V to 4.3 V, Typical 3.3 V
- Power Consumption (GPS +Galileo + IRNSS):
 - Acquisition: 99 mA @3.3 V
 - **Tracking: 95 mA @3.3 V**
- Weight: Approx. **8.2 g**

Other Information

- **Horizontal Position Accuracy**
 - **Autonomous:** < 1.8 m CEP
- **Velocity Accuracy:** < 0.1 m/s

Backup: Ublox Neo M9: tried and tested GPS module for ESP32

Inertial Measurement Unit

The following are the requirements from an IMU -

- 6 axis -> Need tilt for the gyro
- Need accurate tilt support
- Vibration smoothing
- High G support for accurate trajectory calculation

While considering different IMUs we ran into an issue where we had to decide between higher accuracy and faster readout speed.

Different IMUs considered

Name	Axis	Start Time (ms)	Accuracy (m°/s)	Sampling Rate (Hz)	Power consumption (mA)	Inter face	Cost (INR)
BNO055	9	650	100	100*	50 mA (Full operation)	I2C, UART, SPI	1600
MPU-6050	6	100	70	1000	3.9 mA	I2C	300
MPU-9250	9	1	70	4000	4 mA	I2C	800
LSM6DS3	6	1	70	1000	3 mA	I2C, SPI	600
LSM6DS0	6	1	70	1000	3 mA	I2C, SPI	501

* Accuracy varies based on FS setting, normalized with LSB -> taken as 8g accel / 1000 dps (more than 3 rev / s)

* 6 axis -> accelerometer + gyro, 9 axis -> add magnetometer to add more precision

* + -> Exceed requirements

* Max rating -> 2000 dps, 16 g

Selected Module – MPU9250

We have selected the MPU9250 as we need a very fast sensor readout due to the gyro control system

Also, we can use MPU6050 (as a drop-in replacement) if we find difficulty in integrating it.

Reasons

- Good documentation
- Compatibility with ESP32
- Availability of breakout boards [1]
- Low power
- Guide: [ESP32](#)

Interface

- I2C Interface: Max. bit rate up to 400 kbps
- Sampling Rate: 4 kHz

Power and Weight Requirements:

- 3.1 V to 4.3 V, Typical 3.3 V
- 3.9 mA
- Weight: Approx. **10 g**

Backup: MPU6050 (can act as a drop-in replacement if the selected module is unavailable)



Barometric module

The following are the requirements from a Barometric Module -

- Highly accurate pressure measurement so that we can deploy the chute at the right altitude 500m +/- 10 m. ($\Delta p = 1hPa / 10m$)
- We need at least 1 Pa resolution as that is what we have to send back.
- We will also crosscheck speed from the accelerometer with altitude change from the barometer.

Different Barometers considered

Sensors	Accuracy (Pa)	Sampling Rate (Hz)	Current (mA @ 3.3v)	Size (mm ³)	Interface	Cost (INR)
LPS22HB	1	Up to 75	12 μ A (low-power mode)	2.0 x 2.0 x 0.76	I ² C, SPI	~150
BMP388	0.5	Up to 200	3.4 μ A (low-power mode)	2.0 x 2.0 x 0.75	I ² C, SPI	~500
BMP390	1	Up to 100	3.2 μ A (low-power mode)	2.5 x 2.0 x 0.95	I ² C, SPI	~400

* We might also use the SHT40 which gives us precise temperature along with humidity. We are planning to test this as well.

Selected Sensor – **BMP390**

Enough accuracy for our requirements

Manufacturer: Bosch SensorTec

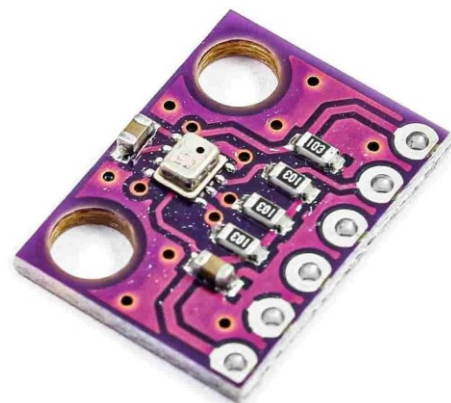
Accuracy: +/- 1 hPa

Power requirements: 2.74 μ @ 3.3V

Ease of use: Guide available [[Guide](#)]

Datasheet: [BMP390](#)

Cost: INR 400 (max)



Temperature sensors

The following are the requirements from a temperature sensor -

- Accurate data to send back to ground station
- Highly accurate so that we can recalibrate barometer to ambient (done automatically for BMP390)

Different modules considered

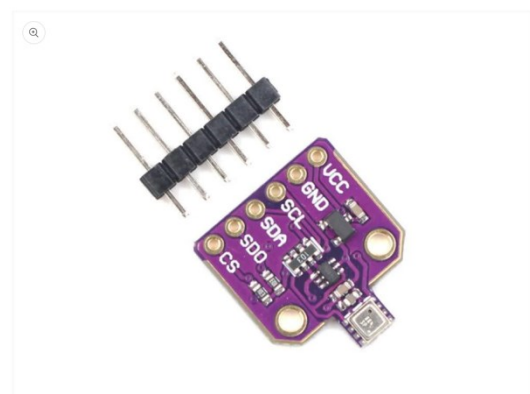
Sensors	Accuracy (°C)	Sampling Rate (Hz)	Current (mA @ 3.3v)	Size (mm ³)	Interface	Cost (INR)
MCP9808	±0.25°C	1	0.1 mA	3 × 3 × 1	I ² C	150
DHT22	±0.5°C	0.5	1.5 mA	15 × 12 × 5	Single-wire Digital	150
BME680	±1°C	1	3 mA	3 × 3 × 0.9	I ² C/SPI	300

Selected Module – **BME680**

The module can provide multiple other readings and can act as a backup for other sensors as well. It also provides good enough reading for our purpose of temperature measurements.

Manufacturer: Bosch SensorTec
Accuracy: ±1°C
Power Requirements: 12 mA @ 3.3V
Ease of Use: Guide Available [[Guide](#)]
Communication: I²C and SPI
Datasheet: [BME680](#)
Cost: INR 750

Backup Module – MCP9808



Humidity sensors

Requirements from a humidity sensor are -

- Stable and reliable operation after a rocket launch
- Fast sensor polling
- Ease of use with ESP32

Different modules considered

Sensors	Accuracy (% RH)	Sampling Rate (Hz)	Current (mA @ 3.3v)	Size (mm ³)	Interface	Cost (INR)
DHT22	5% RH	0.5	~1.5 mA	15 × 12 × 5	Single-wire Digital	150
HDC1080	±3% RH	7	~1 mA	3 × 3 × 0.9	I ² C	200

Selected sensor – HDC1080

Manufacturer: Texas instruments

Accuracy: +/- 1 %RH (relative humidity)

Power requirements: 190 μ @ 3.3V

Ease of use: Guide available [[Guide](#)]

Datasheet: [HDC1080](#)

Cost: INR 400 (max)



Battery management system

We need a battery management system that can tell us current voltage and current so that we monitor battery health and capacity.

Requirements

- Need to measure total current of the system reliably
- We plan to dynamically move sensors offline if current draw nears the max possible.

Different modules considered

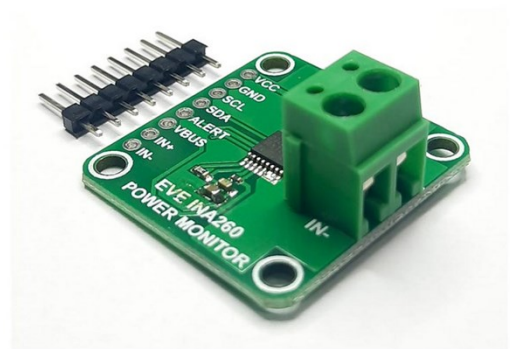
- Using analogue pin on microcontroller
- INA219B (Texas instruments) (Up to 3.2 A)
- **INA260 (measuring more than 3.2A)**

Selected module – INA260

Initially, we will try to use the analogue pin on ESP32 which can read up to 3.3v for the battery system. But once we have the gyro motors adding to the load, we will have to switch to INA260 which can read up to 15A

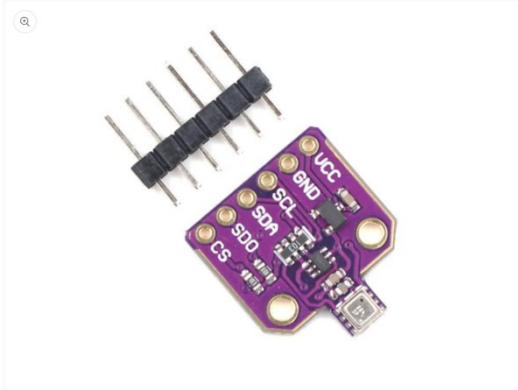
Other Information

- **Manufacturer:** Texas instruments
- **Accuracy:** 10 mA
- **Power requirements:** 310 μ A@ 3.3V
- **Ease of use:** Guide available [[Espressif](#)]
- **Datasheet:** [INA260](#)
- **Cost:** INR 600



Multi Modules

BME680



We plan to use the Bosch sensorTec BME680 sensor which has the following components:

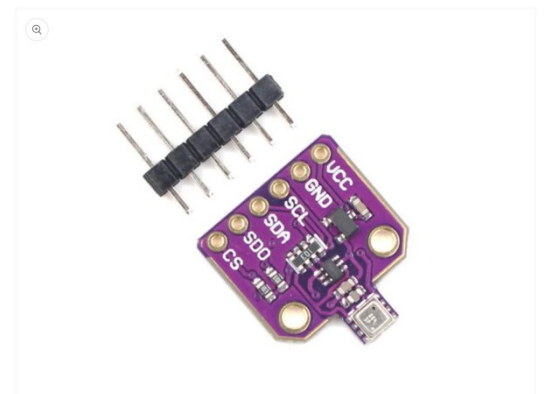
- Relative humidity
- Barometric pressure
- Ambient temperature
- Gas (VOC)

We will use it to perform sensor fusion with different sensor and have a redundant sensor in case of failure on anyone.

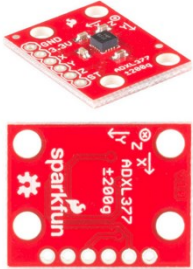
It will also be our main temperature sensor if needed.

Information

- Power: **12 mA @ 3.3V**
- Dimensions: 20 x 16 x 4 mm
- Weight: ~2g
- Ease of use: Guide Available [[Guide](#)]
- Update rate: 1 Hz
- Interface: I²C and SPI
- Cost: 750/-
- Datasheet: [BME680](#)



High G Accelerometer – ADXL377



A high G accelerometer will be helpful in tracking the rocket launch and current location of the launch site with respect to the can sat. This will help us if we plan to implement a return to home system.

As we assumed that the rocket launch will be more than **30g**, which can't be handled by most accelerometers, we found this ADXL 377 based accelerometer which can handle up to **200g**.

Other Information

- **Manufacturer:** Analog Devices
- **Accuracy:** +/- 0.15 g (FS=30g)
- **Power requirements:**
- **Ease of use:** Guide available
- **Cost:** 2500
- **Datasheet:** [Analog devices](#)

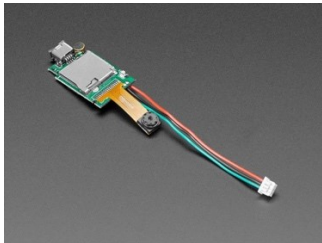
We will only be using this accelerometer if we are able to implement a RTH system.

Camera

Requirements

- Small compact camera
- Onboard SD card recording
- >360 p resolution
- Ability to record far away objects

Modules considered

Feature	RunCam Split V4	Adafruit Spy Camera	ESP32-CAM
Image			
Resolution	1080p @ 60fps	VGA (640x480)	Up to 2 MP (1600x1200)
Lens	140° wide-angle	Standard	Standard
Weight	20g (approx.)	~2g	~9g
Power	5V, 650mA	3.7-5V, low power	5V, low power (~160 mA during use, ~30 mA idle)
Storage	MicroSD up to 128GB	MicroSD up to 32GB	Depends on microcontroller (can stream to a server)
Output	FPV + onboard recording	Onboard recording only	Wi-Fi streaming, supports JPEG image capture
Cost	INR 7000	INR 1000	INR 500
Pros	High-quality 1080p video with wide-angle view	Lightweight and power-efficient	Low cost
	Dual function: FPV and onboard recording	Simple onboard storage	Can stream to a server
Cons	Higher power consumption and weight	Lower resolution and no real-time video feed	Limited onboard storage
	Requires more complex integration		Requires Wi-Fi for streaming

Payload Subsystems Overview

Payload Subsystems Overview

Air Quality Sensor

We are planning to take advantage of the CanSat platform to demonstrate that is possible to conduct atmospheric analysis cheaper with a reusable CanSat.

Dedicated sensors



Particulate matter sensors

We are planning to do check particles of PM 2.5 micron size.

- Honeywell impact 2.5 PM
<https://indusautomation.co/product/honeywell-sensor-pm-2-5/>
- SN-GCJA5
 - <https://www.mouser.in/ProductDetail/Panasonic/SN-GCJA5?qs=XeJtXLI041TL5Ge4ALTvYw%3D%3D>



- **PAS OUT 01** [[Datasheet](#)] (Available in college)
 - Power requirement: 70 mA @ 5V
 - Resolution: 1 $\mu\text{g} / \text{m}^3$
 - Cost: INR 2000
- **SDS011** [[Datasheet](#)] (Available in college)
 - Power requirement: >1W @ 4.7~5.3V

- Cost: INR 3000

Gas concentration sensors

This class of sensor is calibrated to measure the concentration of a specific types of gas. We are picking the gas that have the most difference in under 1 km (NO_x , SO_x and O_3) They are pollution indicators, and we need the following modules to have a reading on them

- MQ135 [[Datasheet](#)] (Available in college)
 - Gas: Smoke
 - Power requirements: 150mA @ 5V
 - Cost: INR 150
- MQ131 [[Datasheet](#)]
 - Gas: Ozone
 - Range: 100 – 10000 ppm
 - Power requirements: ~1W @ 5V
 - Cost: INR 1700

We will test various sensors and pick the most optimal sensors after test at different altitudes.

We plan to check the difference between values at the sixth and

Data produced

- Temperature altitude graph
- Humidity altitude graph
- Pressure altitude graph (location as well)
- PM altitude graph
- Gas concentration vs altitude graph
- Wind speed / direction measurement
- Atmospheric boundary layer height (wind speed)

RIS (Reconfigurable intelligent surfaces)

Our college is working on RIS based antenna for a while now. We are planning to check if we can integrate that technology in our CanSat to achieve the following objectives

- Non-LOS xBee i.e. non holes in exterior
- Live video with xBee or other method

A **Reconfigurable Intelligent Surface (RIS)** [[etsi.org](https://www.etsi.org)] is a programmable surface structure that can be used to control the reflection of electromagnetic (EM) waves by changing the electric and magnetic properties of the surface. These surfaces can be

strategically placed in the radio channel between a transmitter and receiver to control the way the signal reflects off a surface in its propagation path. Reconfigurable Intelligent Surfaces can be used to steer signals to the receiver resulting in better reception or link quality.

We need to make the internal CanSat structure in such way that it becomes a RIS antenna.

Communications and Data Handling Subsystems Design

Communication and Data Handling (CDH) Subsystem Design

Communication Pipeline Overview

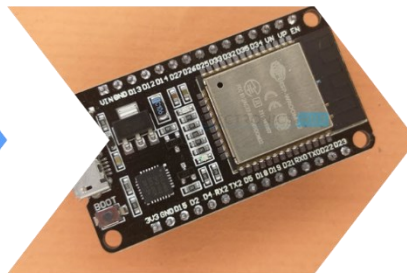
Raw Data from Sensors



Sensor Data

- Temperature
- Pressure
- Humidity
- GPS
- Battery Voltage
- etc

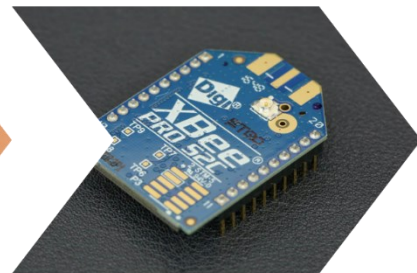
Central Computer System



Computer System

- ESP-32 / Teensy 4
 - Data Processing and Command Executions
- Micro SD Card
 - Data Storage and Backup

Communication to GCS



Communications

- XBEE PRO S2C
 - Radio module
- Antenna
 - Signal Range Extender

Incoming Sensor Data

Onboard the CanSat we have the following sensor data, to analyse, record and transmit to the ground control system:

Sr. No.	Data	Unit
1.	GPS	Latitude (°), longitude (°)
2.	Temperature	Kelvin (°)
3.	Pressure	Pascal (Pa)
4.	Power Measurement (Voltage)	Volts (V)
5.	Humidity	Percentages (%)
6.	Air Quality	Multiple units (ppm, etc.)
7.	Air Speed	m/s ²
8.	Acceleration	m/s ²
9.	Tilt and Rotation	deg/s
10.	Camera	N/A

Central Computer System

Considering the number of devices that must be connected to the central computer system and other factors like boot time, clock speed etc we are considering the following main processing devices. Final deliberation will depend on the performance and stability of each device once all sensors are connected to it. For the boot time, we are considering the minimum possible boot time in the most barebones setup with the smallest possible program.

Reasons for Not Using Microprocessors (Like Pi Zero)

The reason we decided to use only microcontrollers is because the code is being run directly on the CPU without an OS in between. Adding an OS would include other overheads that can potentially increase boot time and would need to get startup codes that autorun our code on startup.

This can be a nightmare in case the code fails, and we must reboot the processor wherein the increase load times can lead to catastrophic consequences regarding gyroscopic control and the control of the secondary descent mechanism.

Requirements from the Central Computers

Following parameters were considered are as follows –

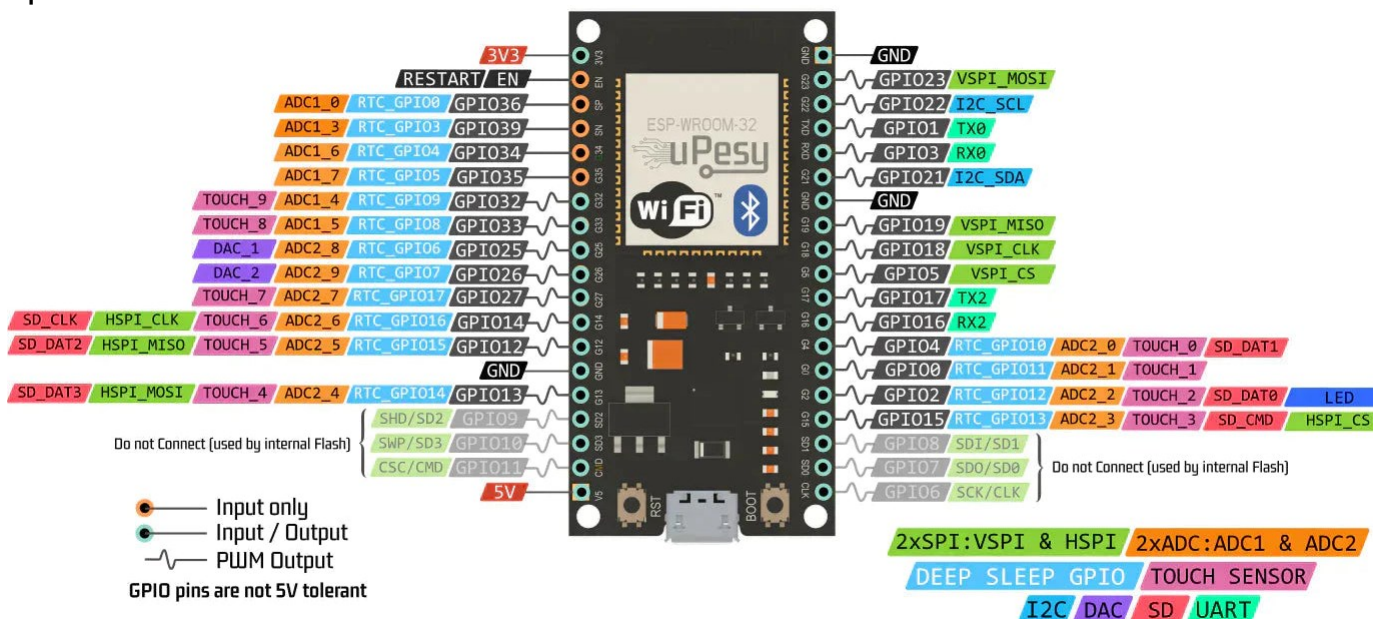
- Processor Speed
- Memory
- Interfaces available
- Input Voltage
- Cost

Different Central Computers Considered

Sr No	Name	Processor	Processor Speed (MHz)	Memory (kB)	Data Interfaces	Input (V)	Op. Temp (°C)	Cost (INR)
1.	Raspberry Pi Pico 2	Dual-Core ARM Cortex M33	150	520 kB SRAM	GPIO (x26) UART (x2) SPI (x2) I2C (x2) PWM (x24) Analog (x3)	1.8-5.5	-20 to 85	600
2.	Teensy 4.1	ARM Cortex M7	600	7936K Flash, 1024K RAM, 4K EEPROM	GPIO (x55) UART (x8) SPI (x3) I2C (x3) PWM (x35) Analog (x18)	5	-40 to 85	2700
3.	ESP32-WROOM-32E1	Xtensa dual-core 32-bit LX6	80 - 240	448 KB ROM, 520 KB SRAM and 16KB SRAM in RTC	GPIO (x38) UART (x2) SPI (x2) I2C (x2) PWM (x23) Analog (x16)	3.6	-40 to 85	1700

Selected Processing Modules

We have decided to select the **ESP32-WROOM-32E** as the main processing computer. With Teensy kept as a backup which may be used but rigorous testing is required.



PIN OUT OF ESP-32

Reasons for Deciding to Use ESP32

Following are the reasons for selection of ESP-32

- Powerful Dual Core Processor
- High Clock Speed
- Sufficient pins for Sensors planned to use
- Ease in Trouble Shooting for errors
- Existence of plentiful guides on the Internet
- Can be stripped down to have a chipset for PCB
- Ease of coding and availability of Libraries

The ESP32 is also readily available. In case the exact ESP32-WROOM-32E is not available we can still go forward with other versions like the ESP-32-WROOM-32UE/D/... all of which have similar configurations, pinout and features which we can use for our build.

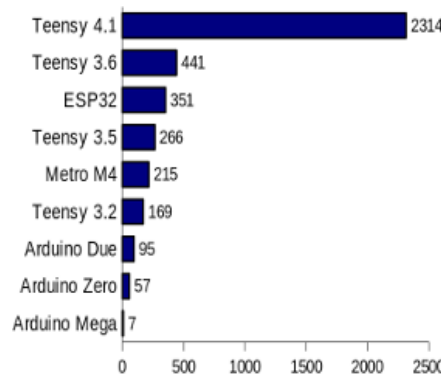
One of the cons with using the ESP32 is the fact that we must add extra hats for adding an SD card, a feature that is already inbuilt in all the other microcontrollers we considered.

Other Boards Being Considered

The Teensy 4.1 is also being considered as a central processing computer for our CanSat



Teensy 4.1 has a very powerful processor and more GPIO, UART, SPI and other pins which improves connectivity. It also has an SD card slot built into it which removes the need for a separate board just to interface an SD card for storage. Everything except the bootloader of the board is open source which gives us a lot of power to modify with the board as we see fit.



The only problem with Teensy is that it is not a widely used processor and therefore any issues that we come up with during the development of the system can be too hard for even us to solve. A lot of third-party extension boards do not exist for the Teensy 4.1 and sensor compatibility can be an issue. Although the board is supposed to be Arduino compatible and can be programmed using the Arduino IDE, to fully utilize the board we may have to use other platforms like Visual Micro, Platform IO or Circuit Python.

Still, we want to try using the Teensy for our project due to the big performance boost that it provides. Therefore, the Teensy 4.1 is still a potential board that we can use for our CanSat, provided we can do proper integration testing with all our sensors.

Central Computer System: A Summary

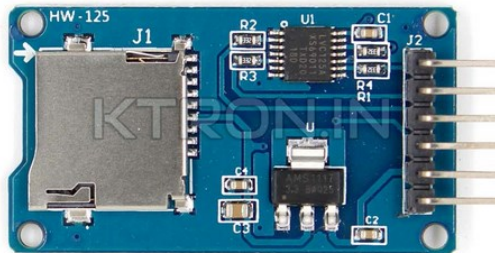
These are the two microcontrollers being considered:

1. ESP32-WROOM-32E (Currently selected)
2. Teensy 4.1 (Will only select after rigorous integration checks with all sensors and other modules)

Payload Memory

Provided the fact that we are as of now going forward with an ESP32 Module, we will have to incorporate a third-party board to attach an SD card module to the microcontroller.

We can attach an SPI based SD Card Module from KTRON.



The SD Card module has the following features:

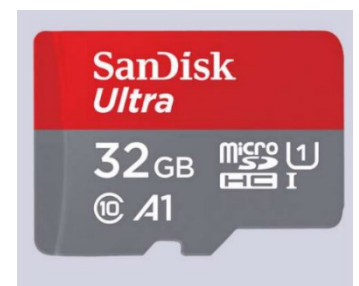
1. Operating voltage: 4.5 to 5V
2. Communication interface: SPI
3. 4 x M2 screws
4. Support Micro SD Card ($\leq 2G$), Micro SDHC card ($\leq 32G$) (high-speed card)

SD Card Selections –

Sr No	Name	Memory (GB)	Data read Speed (Mbps)	Speed Class	Cost (INR)
1.	SanDisk Ultra microSD UHS-I Card	32	120	10	369
2.	SanDisk High Endurance Micro SDHC Card	32	100	10	729

Deciding to Use UHS-I

1. Faster data read and write speeds.
2. Readily available
3. High Endurance is for long time usage (Because we are only using the SD card for the duration of the flight of CANSAT, the increase cost does not justify the configuration)



Payload Antenna

For increasing the connectivity range, for the XBEE we decided to go forward with a custom antenna. We got the following antenna with the specifications:

Sr. No.	Name	Gain (dB)	Operating Band (GHz)	Type	Cost (INR)
1.	3AN High Gain 12 dBi Wi-Fi Router Antenna	12	2.4-5.8	Omni-Directional	499
2.	XBEE S2C with Wire Antenna	1.5	2.4	Omni-Directional	499

Selected Antenna: High Gain Wifi Router Antenna (highlighted in green):

1. Better gain
2. XBEE S2C with inbuilt antenna had lesser range as compared to using the XBEE Pro S2C



Payload Radio

Following are the Radios we considered for CANSAT –

Sr. No.	Name	Operating Frequency (GHz)	Transmit Current (mA @ 3.3 VDC)	Receive Current (mA @ 3.3 VDC)	Transmit Power (mW)	RF LOS Range (km)	Cost (INR)
1.	XBEE PRO S2C	2.4	120	31	63 (+18 dBm)	3.2	2000
2.	XBEE PRO 900 HP (XBEE PRO S3B)	0.9	229	44	Up to 250mW (+24 dBm)	15.5	4700

We already have an XBEE PRO S2C in our inventory which has been tested already. Given the fact that most of the XBEE chips are out of stock in major electronics parts retailers, this is a good option for us.

The newer versions of the chip are the XBEE RR PRO and the XBEE 3 PRO both of which give same range and data transfer speeds of 250 Kbps for RF. Therefore, we are considering them too depending on availability and cost.

Another radio we are considering us is the XBEE PRO 900 HP which can provide a (claimed) range of 28km line of sight (with high gain antenna). Depending on availability and after checking the true range of the chip we may choose to switch to it later (hence highlighted by orange).



XBEE RR PRO



XBEE 3 PRO



XBEE S2C

Alternative Technologies for Data Transfer

We are also considering the use of LoRa WAN as a data transmission method in case we the XBEE is not able to work properly. LoRa has long range capacity and can be designed for low-power applications. However, data rate can become a problem as we must maintain a steady data rate of 1Hz with all the data that we have to send from the Flight Software to the Ground Control Software.

We are planning to work with the XBEE radio itself. However, in case the XBEE does not give us satisfactory range during integration and overall systems testing we will be using a lora module instead.

GPRS/GSM

We will be used a very widely available SIM800L module to transmit data using 2G to our server.

Product link => [SIM800L](#)
[[Product Link](#)] => [Product Link](#)

This will act as a backup data transmitter alongside the LoRa/Xbee primary transmitter. However, this has a peak power draw of 2 Amps which is more likely to be achieved given the height, and a working voltage of 3.7 to 4.2 volts. Now this means that we either need a dedicated voltage regulator for this, but at that peak draw it can significantly drain the battery. The other alternative solution is to use a singular 18650 dedicated for this, whose output voltage matches that of the required input. If the dedicated battery depletes then this system simply stops working without affecting any other components.

Note that we are only planning to use the voltage regulator right now but based on further testing we might plan to use the additional battery for safety.

We want the system to be able to sustain much more than the required operation time instead of cutting it close.

This can also give us its location using triangulation techniques (there are easily available commands that do this). Useful in beacon mode as well.

We plan to send custom UDP packets using this to a cloud server accessible on the public internet.

This sends data in the same format as the primary transmission and with the same interval as well.

However if for some reason the cansat stays in air for much longer than anticipated and below a certain battery voltage threshold this will be deactivated to save energy.

Payload Radio Configuration

1. The NETID/PANID of both the Payload and the GCS XBEE devices will be set to the TEAM ID
2. The XBEE devices are configured to communicate with each other by addressing the specific device, thus disabling broadcast mode.
3. The GCS XBEE will be set as the Coordinator and the Payload XBEE will be set as the End device.
4. At each mission phase, the payload transmits the telemetry to the GCS at a rate of 1Hz.

Payload Telemetry Format

1. Telemetry is transmitted at a rate of 1Hz
2. The format is as follows:

<TEAM_ID>, <TIME_STAMPING>, <PACKET_COUNT>, <ALTITUDE>,
<PRESSURE>, <TEMPERATURE>, <VOLTAGE>, <GNSS_TIME>,
<GNSS_LATITUDE>, <GNSS_LONGITUDE>, <GNSS_ALTITUDE>, <GNSS_SATS>,
<ACCELEROMETER_DATA>, <GYRO_SPIN_RATE>,
<FLIGHT_SOFTWARE_STATE>, <MODE>, <AIR_SPEED>,
<AIR_QUALITY_DATA>, <HUMIDITY>, <AEROBREAK_DEPLOYED>,
<CMD_ECHO>

Parameter name	Format	Description
<TEAM_ID>	2024ASI-CANSAT-XXX	The Team number of the team
<TIME_STAMPING>	ss	Time in seconds since the initial power
<PACKET_COUNT>	N	Total count of transmitted packets since turn on (maintained through processor reset)
<ALTITUDE>	N.N	Altitude in meters (relative to the ground level at the launch site) with a resolution of 0.1 metres
<PRESSURE>	N	Air pressure if the sensor used with a resolution of 1Pa
<TEMPERATURE>	N.N	Temperature in degrees Celsius with a resolution of 0.1 degrees
<VOLTAGE>	N.NN	Voltage of the CanSat power bus with a resolution of 0.01Volts
<GNSS_TIME>	hh:mm:ss	Time from the GNSS receiver with a resolution of 1 second
<GNSS_LATITUDE>	N.NNNN	Latitude from the GNSS receiver with a resolution of 0.0001 degrees
<GNSS_LONGITUDE>	N.NNNN	Longitude from the GNSS receiver with a resolution of 0.0001 degrees
<GNSS_ALTITUDE>	N.N	Altitude from the GNSS receiver in metres above mean sea level with a resolution of 0.1 metres.
<GNSS_SATS>	N	Number of GNSS satellites connected to the receiver
<ACCELEROMETER_DATA>	N,N,N	Data received from gyroscopic sensor, i.e. roll and pitch parameters in m/s^2
<GYRO_SPIN_RATE>	N,N,N	Spin rate of the Mechanical Gyro with respect to the CanSat in deg/s

<FLIGHT_SOFTWARE_STATE>	string	Operating state of the software which includes the following: BOOT, TEST_MODE, LAUNCH_PAD, ASCENT, ROCKET_DEPLOY, DESCENT, AEROBREAK_RELEASE and IMPACT
<MODE>	F or S	'F' for flight mode and 'S' for simulation mode
<AIR_SPEED>	Floating point	Air speed in m/s (measured with the pilot tube)
<AIR_QUALITY_DATA>	N,N,N,...	Air quality data from multiple sensors
<HUMIDITY>	N	Humidity value from sensors
<AEROBREAK_DEPLOYED>	Y or N	Y indicated that the aero break (secondary descent control mechanics) is deployed. N indicates that it has not been deployed.
<CMD_ECHO>	string	Text of the last command, received and processed by the CanSat

Payload Command Formats

This is a list of commands that can be used to control different aspects of the CanSat from the Ground control system itself.

1. CMD, <TEAM_ID>, TX, <ON/OFF>: CX (Payload Telemetry ON/OFF people)
 - a. CMD and TX are static text
 - b. <TEAM_ID> is the assigned team identification
 - c. <ON/OFF> is the string 'ON' to activate the payload telemetry transmission and 'OFF' to turn off the transmissions.
 - d. Example: The command CMD, 2024ASI-CANSAT-XXX, CX, ON activated the transmission of payload telemetry
2. CMD, <TEAM_ID>, ST, <UTC_TIME> | GNSS: ST (Set time)
 - a. CMD and ST are static text.
 - b. <TEAM_ID> is the assigned team identification.
 - c. <UTC_TIME>|GNSS is the date in the format:
 - i. UTC time in the format hh:mm:ss

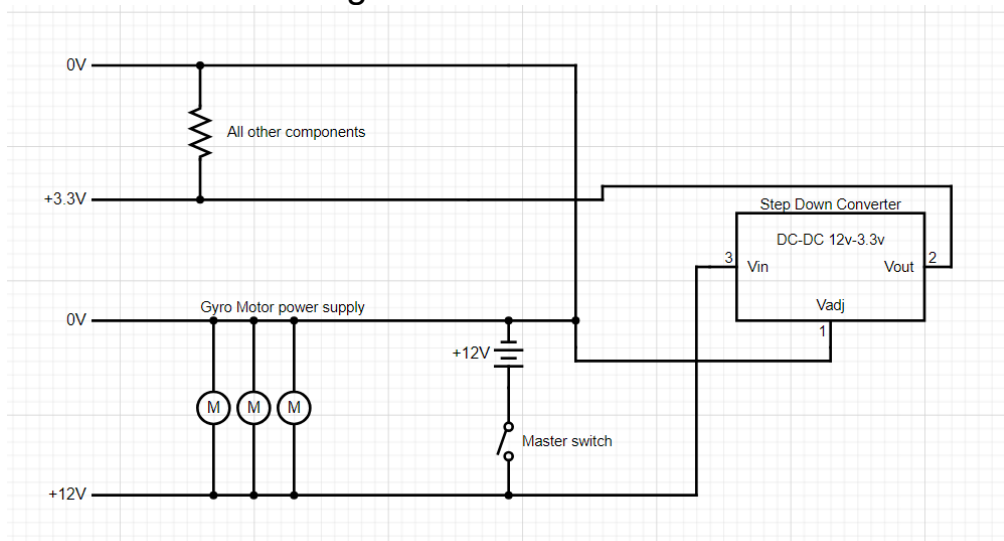
- ii. GNSS which sets the flight software time to the current time read from the GNSS module.
 - d. Example: The command CMD, 2024ASI-CANSAT-XXX, ST, <TIME> updated the time and set the time to current GPS time from the GPS sensor.
3. CMD, <TEAM_ID>, SIM, <MODE>
- a. CMD and SIM are static text
 - b. <TEAM_ID> is the assigned from identification
 - c. This will also add Test Mode in <FLIGHT_SOFTWARE_STATE>
 - d. <MODE> is a string:
 - i. 'ENABLE' to enable the simulation mode
 - ii. 'ACTIVATE' to activate the simulation mode
 - iii. 'DISABLE' which both disables and deactivates the simulation mode.
 - e. CMD, 1000, SIM, ENABLE will enable simulation mode and change flight software state to TEST MODE.
4. CMD, <TEAM_ID>, SIMD, <DATA>, <FIELD>
- a. CMD and SIMD are static text
 - b. If the field is valid, it sends the simulated data to the payload.
 - c. Will only work if the whole CanSat is in TEST_MODE
5. CMD, <TEAM_ID>, ALT_CAL
- a. Used to calibrate the altitude to 0 when the CanSat is waiting at the launch pad as the altitude of the CanSat is the height from the launchpad.
6. CMD, <TEAM_ID>, BCN, <ON/OFF>
- a. CMD and BCN are static text
 - b. <ON/OFF> allow to test the audio beacon which would otherwise only turn on upon landing
 - c. Used for testing purposes.

Electrical Power Subsystem Design

Electrical Power Subsystem (EPS) Design

#	Name	Quantity	Voltage (V)	Capacity (mAh)	Weight (g)	Volume (mm*mm*mm)	Capacity / Weight ratio	Cost (INR)
1	18650	3	$3.7 \times 3 = 11.1$	2200	$50 \times 3 = 150$	18*18*65	14.66	$245 \times 3 = 735$
2	21700	3	$3.7 \times 3 = 11.1$	4200	$68 \times 3 = 204$	21*21*70	20.58	$510 \times 3 = 1530$
3	18650 3C pack	1	11.1	2200	184	70*65*25	11.95	949
4	21700 3C pack	1	11.1	5000	271	120*100*30	18.45	1849

Even though using three 18650 cells gives us a better capacity to weight ratio, the 3C pack is better as it contains the protection circuits as well. 21700 batteries are better in general but are more expensive and the weight is not justifiable for our use case. Still, it is something we could consider in the future.



Variable Converter This has a high max load current of 5A, we also have option to use 16v 4s1p config if we need to later

Fixed converter This has a max current output of 3.3A at 3.3V

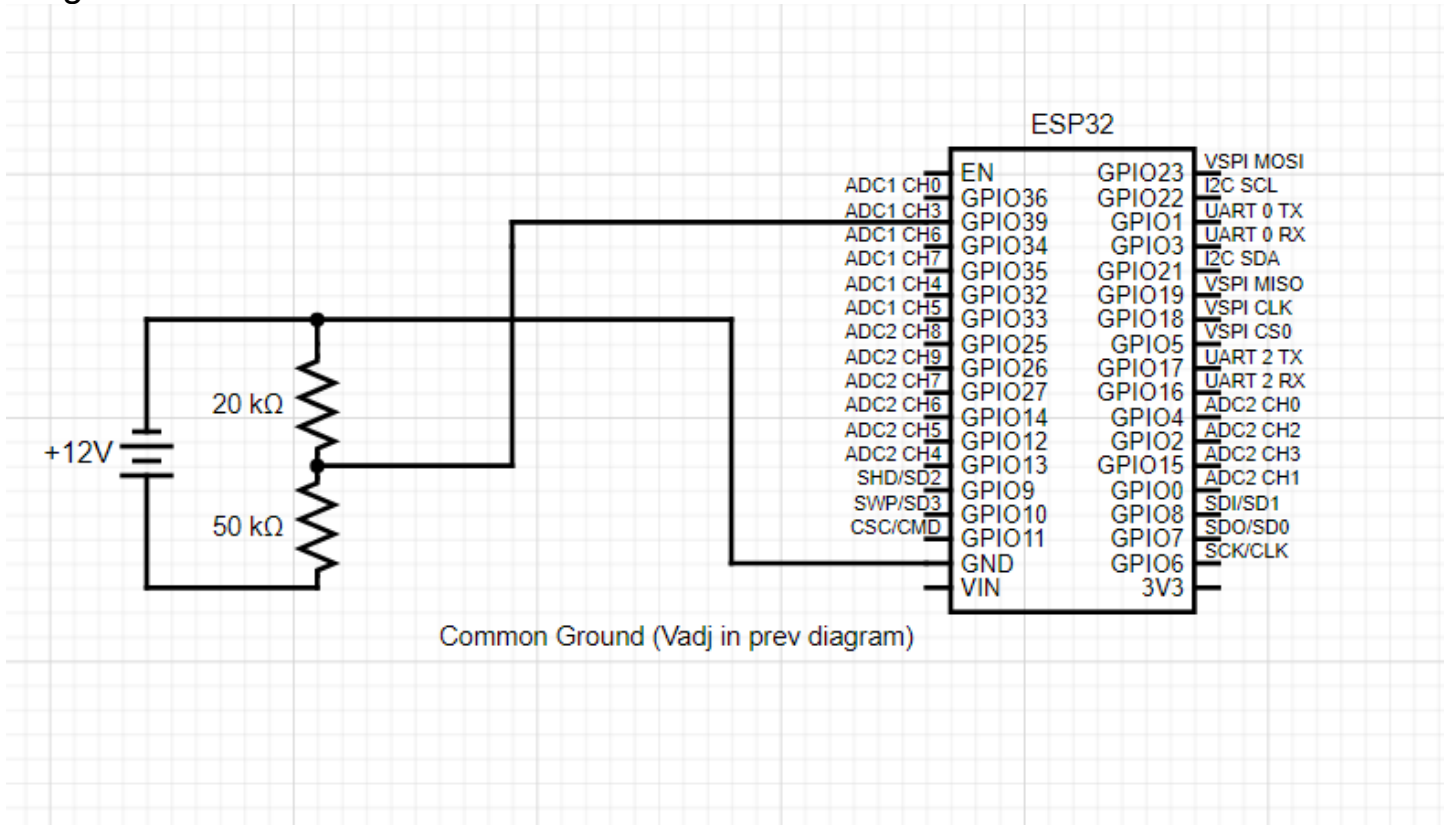
Both converters have good efficiency and are ideal for our usage, although first one has better performance on paper and definitely more flexibility.

As soon as master switch is opened (OFF position), all components are shut down.

Measuring the battery voltage

The voltage regulator is very accurate maintains 3.3V with minimal error, given this fact we can calculate the potential difference of the battery using analogue Read on the esp32 itself. But the error in this calculation process will be a function of the errors of the 2 resistors used, the esp32 analog channel, and the step-down converter.

Diagram:



$$V_{batt} = \left(\frac{x}{4096} \right) * \left(\frac{20 + 50}{20} \right) * 3.3$$

Where 4096 is the default resolution of analogue Read in esp32, x is the value obtained after doing analog Read.

Total Power usage of the Cansat is given in the [Power Budget](#) section.

Flight Software Design

Flight Software (FSW) Design

General

- Programming Language – C/C++
- Development Environment – Visual Studio Code with Platform I/O extension for Teensy 4.1 and Arduino IDE extension for ESP32.

We are trying to achieve the following objectives using the FSW: -

1. Collect raw data and convert it into required units with required precision.
2. Store the data on-board using an SD card and to make a backup of the FSW state regularly so that the Central Processing Unit can load back to previous state
3. Send the sensor data to Ground station via XBEE transmission and other methods of data transfer.
4. Receive commands from Ground Station to perform calibration, changing mode, setting time, etc [Check Payload Command for more information.]
5. Initiate the release of the secondary descent control mechanism at 'x' metres.
6. Finely control the movements of the 3-axis gyroscope present inside the CanSat.
7. To keep track of transmitted packets and mission time despite processor resets.

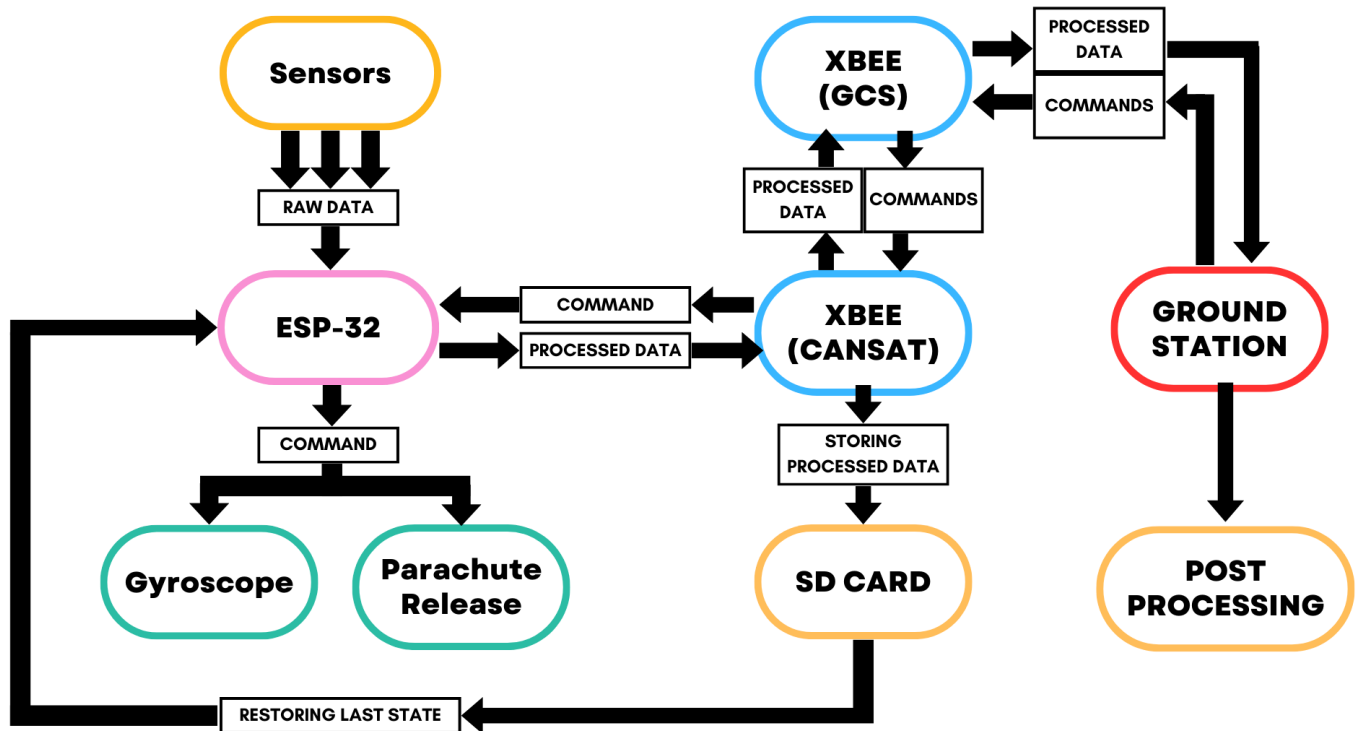
Simulation Software:

1. Taking data from the ground control station and provide output depending on the altitude, pressure and other values to check if all functions are working as required (but on ground)

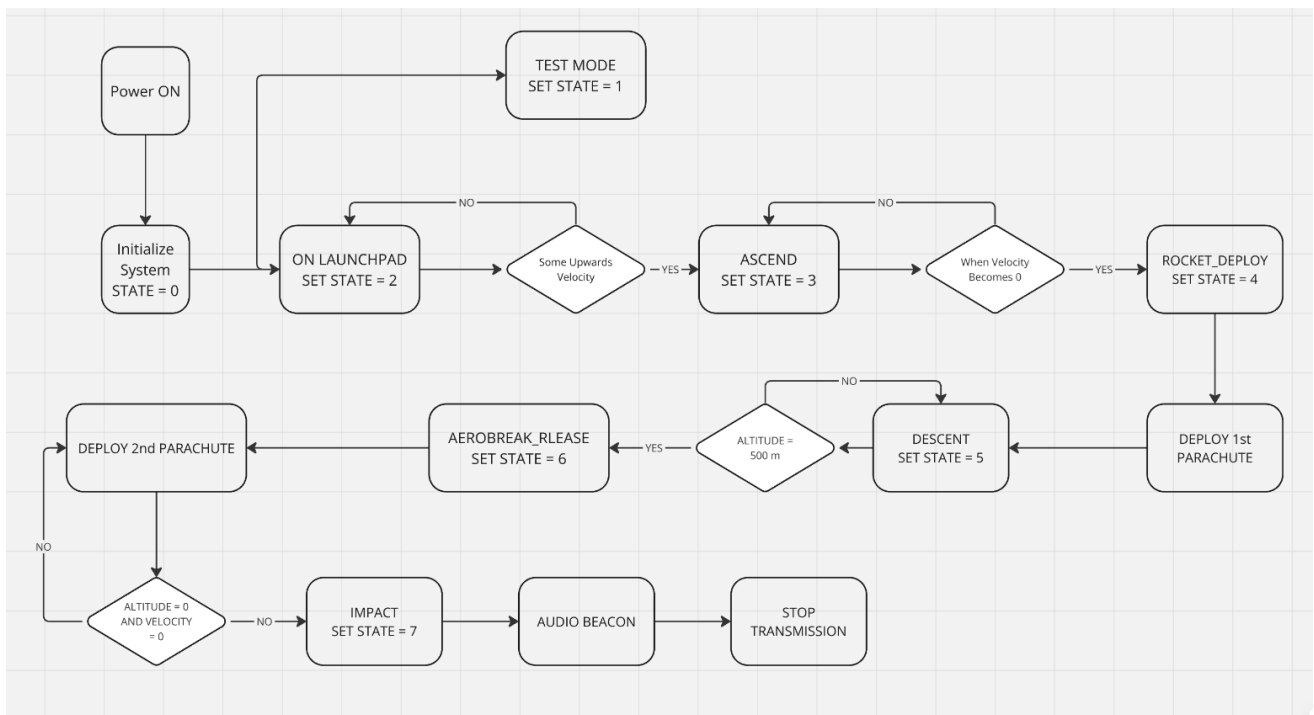
Flight Mode:

1. Upon power on, the flight systems will be initialized.
2. Upon receiving the Set Time command from the GCS, the CanSat time will be set within 1 second of UTC (or the mentioned time) prior to the launch.
3. At the launchpad, when the FSW received the altitude control command, the altitude will be calibrated to 0m.
4. Sampling rate of sensors will be at their maximum possible value (to ensure that the altitude measurement is accurate for parachute and heat shield deployment).
5. Data will be sent to the XBEE at a polling rate of 1Hz

FSW and GCS Connection Diagram



FSW State Diagram



During Test Mode, we follow the same state diagram, the only difference being that the values are simulated by the Ground Control System.

Processor Resets

Data used for handling processor reset:

1. Mission time, altitude calibration data, number of packets transmitted, most recent command received, flight mode, parachute status, telemetry status are store in SD card in CSV format.
2. This data can be recovered from the SD Card in case of processor reset or power loss.
3. In case of processor reset, we can use these data to resume flight operations such as zero altitude calibration.

Possible reasons for processor resets:

1. Temporary power failure due to disconnections.
2. Software bugs
3. Overheating of microcontroller
4. Power fluctuations to the microcontroller

Simulation Software

This is the list of commands available for simulation mode:

1. CMD, <TEAM_ID>, SIM, <MODE>
 - a. CMD and SIM are static text
 - b. <TEAM_ID> is the assigned from identification
 - c. This will also add Test Mode in <FLIGHT_SOFTWARE_STATE>
 - d. <MODE> is a string:
 - i. 'ENABLE' to enable the simulation mode
 - ii. 'ACTIVATE' to activate the simulation mode
 - iii. 'DISABLE' which both disables and deactivates the simulation mode.
 - e. CMD, 1000, SIM, ENABLE will enable simulation mode and change flight software state to TEST MODE.
2. CMD, <TEAM_ID>, SIMD, <DATA>, <FIELD>
 - a. CMD and SIMD are static text
 - b. If the field is valid, it sends the simulated data to the payload.
 - c. Will only work if the whole CanSat is in TEST_MODE

Other Commands Offered By FSW

1. CMD, <TEAM_ID>, TX, <ON/OFF>: CX (Payload Telemetry ON/OFF people)
 - a. CMD and TX are static text
 - b. <TEAM_ID> is the assigned team identification
 - c. <ON/OFF> is the string 'ON' to activate the payload telemetry transmission and 'OFF' to turn of the transmissions.
 - d. Example: The command CMD, 2024ASI-CANSAT-XXX, CX, ON activated the transmission of payload telemetry
2. CMD, <TEAM_ID>, ST, <UTC_TIME> | GNSS: ST (Set time)
 - a. CMD and ST are static text.
 - b. <TEAM_ID> is the assigned team identification.
 - c. <UTC_TIME>|GNSS is the date in the format:
 - i. UTC time in the format hh:mm:ss
 - ii. GNSS which sets the flight software time to the current time read from the GNSS module.
 - d. Example: The command CMD, 2024ASI-CANSAT-XXX, ST, <TIME> updated the time and set the time to current GPS time from the GPS sensor.
3. CMD, <TEAM_ID>, ALT_CAL
 - a. Used to calibrate the altitude to 0 when the CanSat is waiting at the launch pad as the altitude of the CanSat is the height from the launchpad.
4. CMD, <TEAM_ID>, BCN, <ON/OFF>
 - a. CMD and BCN are static text
 - b. <ON/OFF> allow to test the audio beacon which would otherwise only turn on upon landing
 - c. Used for testing purposes.

Software Development Plan

1. Each sensor is tested individually by supplying their operating voltage.
2. A prototype of the payload is created by connecting all components and supplying voltage. The output is verified by comparing to previous individual results.
3. Time required for booting of microprocessor is calculated by measuring time interval between a series of blinking signals.

Ground Control System

Ground Control Software

Programming Language: python

We will create a simple script running on a machine connected to an Arduino that captures and stores all data received by the XBEE



A python program running on the machine will read all serial input from Arduino continuously and keep appending the packets (after formatting) to a csv file. For Visualization purposes we will build a very simple app that uses the csv file and renders it on a browser.

Power considerations

Most laptops can easily output up to 10W, and our XBEE will need 4W max. We will still test with a separate battery in case there is any chance of performance gain.

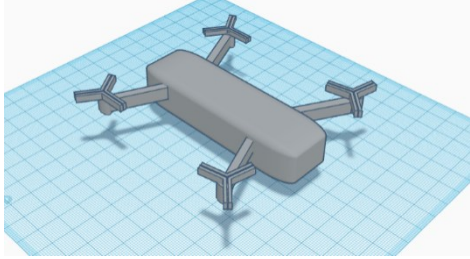
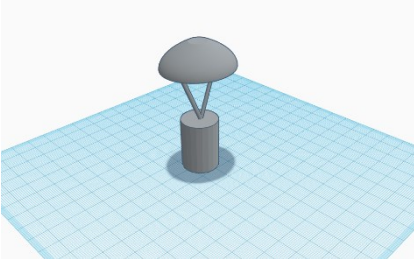
Commands

Commands will be written in the python terminal which will get converted to the appropriate format and then sent to the Arduino which will transmit it using the XBEE.

Descent Control Design

Descent Control Design

For controlling the descent rate of the CANSAT, we have two options, it can either be vertical or horizontal, below are the trade-offs:

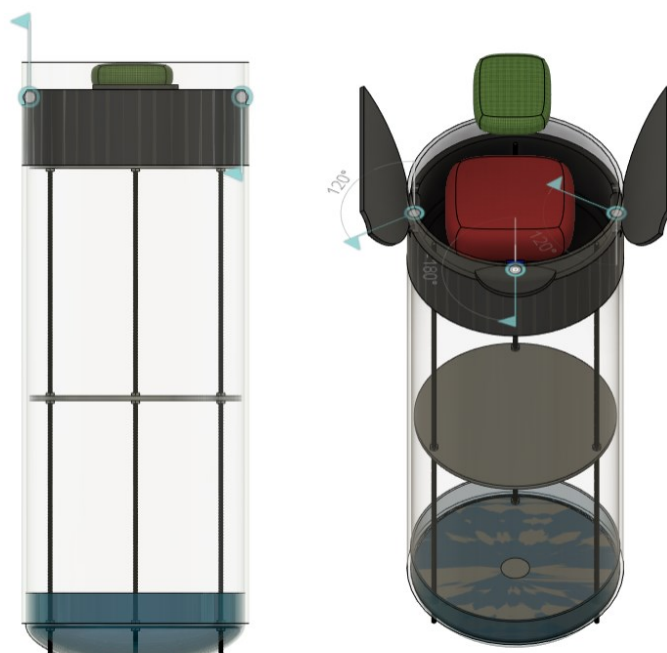
	Horizontal	Vertical
Images for Reference		
Reliability	Moderately reliable: as it depends on aerodynamic factors like wind conditions and the effectiveness of the control surfaces. Precision in descent is affected by external conditions.	Highly reliable: as it focuses on controlling the rate of descent rather than external factors like lateral movement. More consistent performance in various conditions.
Pros	• Improved Directional Control • Increased Descent Distance • Flexibility in Landing	• Highly reliable, as it focuses on controlling the rate of descent rather than external factors like lateral movement. • More consistent performance in various conditions.
Cons	• Affected by Wind: Strong winds or unpredictable atmospheric conditions can greatly impact the horizontal control system • Complex Mechanism: Requires precise control surfaces (like ailerons or wings) and aerodynamic stability,	• Limited Directional Control: Vertical control does not allow for horizontal manoeuvring, making it harder to adjust the CanSat's landing zone or avoid obstacles. • Passive Control Limitations: Depending on passive drag mechanisms can lead to less adaptability

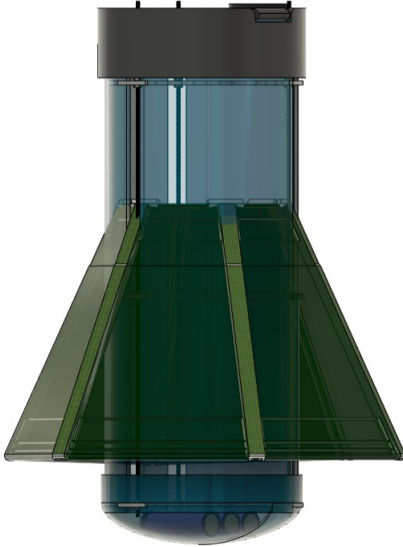
	adding mechanical complexity and potential failure points. • Energy Consumption: Active systems (e.g., motors for controlling flaps or surfaces) can consume additional power, reducing battery life for other mission-critical tasks.	in varying atmospheric conditions, making precision landing difficult.
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Selected Design	Rationale
Vertical	Simplicity and Reduced Mechanical Complexity
	Higher Reliability in Diverse Conditions- Vertical control provides a more stable and consistent descent, unaffected by wind and other external factors that can complicate horizontal control.
	Conservation of Energy: By focusing on passive or low-energy descent methods (like parachutes or flaps), vertical control conserves battery life, which can be used for other mission-critical tasks, unlike horizontal control which may require more active systems.

For the Vertical Landing the CanSat contains a total of 2 Descent Control Mechanisms – Primary Deployment mechanism which is a parachute will be deployed immediately after ejection from the rocket, the Secondary Descent Control Mechanism is deployed at an altitude of 500m and below are few trade-offs for this mechanism:

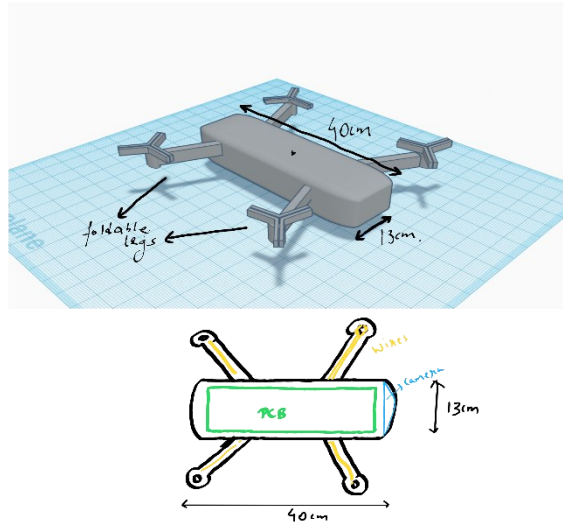
1. Secondary Parachute
2. Umbrella model
3. 4 Rotor model
4. Wings model

Secondary Parachute Design	
Images for reference	
Reliability	The system is generally reliable as the lid mechanism for releasing the second parachute ensures controlled deployment. However, the reliance on the first parachute being deployed correctly from the beginning could introduce some dependency-related risks. If the first parachute doesn't function as expected, it may hinder the secondary parachute's release.
Pros	• Controlled Lid Opening Mechanism: The lid mechanism can ensure the parachute is deployed smoothly and at the right time, reducing the chance of the parachute getting tangled or failing to open fully. • Sequential Deployment: The connection between the first and second parachutes provides a step-by-step deployment sequence, ensuring a smoother descent by gradually increasing drag at different stages. This helps prevent abrupt changes in descent speed.
Cons	• Mechanical Failure Risk: The lid and spring mechanism introduce moving parts that may be prone to mechanical failures, especially if not properly tested or if exposed to extreme environmental conditions like dust or humidity. • Dependency on First Parachute: Since the second parachute is connected to the first, if the first parachute encounters issues (e.g., tangling, improper deployment), it could compromise the activation and performance of the second parachute.

Umbrella Design	
Images for reference	
Reliability	Moderate to High Reliability: The combination of a parachute and inverted umbrella provides redundancy in drag generation. The umbrella acts as a backup to increase drag if needed, improving the overall control of the descent. However, mechanical reliability is important for the spring deployment system.
Pros	• Increased Drag: The inverted umbrella design enhances drag by increasing the surface area exposed to air, ensuring a slow and controlled descent in Stage 2. • Stable Descent: The umbrella provides added stability by evenly distributing the drag forces around the CanSat, reducing the risk of tumbling or erratic movement. • Compact Design: The umbrella can be neatly folded within the CanSat's body and deployed smoothly, ensuring space efficiency without impacting other systems.
Cons	• Mechanical Complexity: The servo motor and spring mechanism for deploying the umbrella add complexity to the design, which could be prone to mechanical failure if not properly tested or maintained. • Wind Sensitivity: The large surface area of the umbrella might make the system more susceptible to strong winds or turbulence, potentially causing instability or difficulty controlling the descent. • Risk of Malfunction: If the umbrella fails to deploy or deploys incorrectly, it could result in an imbalance during descent, increasing the risk of a hard landing.

Rotor Design

Images for reference



Reliability

Moderate Reliability:

While rotors provide fine control over the descent, they rely heavily on the onboard power supply and functioning of all motors. Any malfunction in the rotors, power supply, or control system can lead to an unstable or uncontrolled descent.

Pros

- **Active Descent Control:** Rotors allow for precise control of the CanSat's descent speed, making it possible to adjust in real-time based on wind conditions or obstacles, leading to a smoother landing.
- **Adaptability:** Unlike passive systems, rotors can actively compensate for changes in wind speed and direction, giving the CanSat the ability to adapt to a variety of atmospheric conditions.
- **Potential for Hovering:** In case of uneven terrain, rotors can slow the descent to a hover before landing, providing greater control over the final landing spot.

Cons

- **High Power Consumption:** Rotors require a significant amount of power to operate, especially during the descent phase, which could limit the CanSat's battery life and the ability to run other systems.
- **Increased Complexity and Weight:** Adding rotors introduces additional weight and mechanical complexity, which could lead to higher chances of failure. The more moving parts involved, the more potential points of failure.
- **Higher Risk of Malfunction:** If any of the motors or control systems fail, the descent could become unsteady or uncontrollable, potentially resulting in a hard or imbalanced landing.

Wings Design

Images for reference



Reliability

Moderate Reliability: The passive deployment system minimizes power use, reducing failure risks. However, the addition of mechanical parts like servos and springs adds complexity, and performance can be affected by wind conditions.

Pros

- **Simple and Reliable Deployment:** The use of a spring mechanism for wing deployment minimizes complexity compared to powered systems, making the deployment process more reliable with fewer chances of mechanical failure.
- **Minimized Landing Impact:** By increasing drag through both the parachute and wings, the landing speed is reduced, minimizing the risk of damage to onboard instruments upon touchdown.
- **Reduced Power Consumption:** The system relies primarily on passive elements (parachute and spring-loaded wings), reducing the need for active power usage and conserving battery life for other mission-critical operations.

Cons

- **Increased Structural Complexity:** Adding deployable wings requires additional mechanical components, such as servos and springs, which could increase the risk of mechanical failure and the overall weight of the CanSat.
- **Space Constraints:** The wings need to be folded during initial deployment, which could limit the available space for other components or payloads within the CanSat.
- **Sensitivity to Wind Conditions:** The passive nature of the wings means that their performance can be significantly affected by external factors like wind speed and direction,

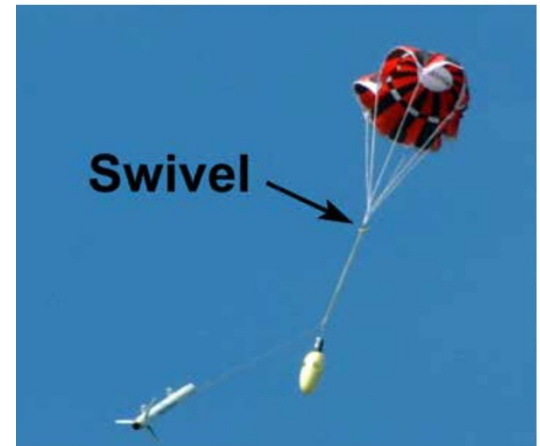
Selected Design	Rationale
Secondary parachute design	Simplicity and Reliability: The secondary parachute design is simpler compared to the rotor and umbrella mechanisms.
	Low Power Consumption: Unlike the rotor-based design, the parachute system requires no additional power for operation during the descent
	Balanced Descent Stability: The secondary parachute enhances stability during descent by distributing drag evenly across the CanSat's structure, reducing the risk of tumbling or erratic movement.

Selected descent control mechanism

Two Parachute (Twin chute) method

Additionally, both the chutes are attached to a swivel to prevent reduction in diameter of chute due to rotation of the CanSat.

- In essence, the roll of the CanSat body and chute gets decoupled and they can rotate independently of each other.



Equations -

$$A = \frac{2mg}{\rho v^2 C_d}$$

$$m = 1\text{kg}$$

$$g = 10\text{m/s}^2$$

$$\rho = 1.22 \text{ kg /m}^3 \text{ at sea level and standard temperature}$$

Mass of parachutes

$$\text{Mass} = \text{Area} \times \text{mass per unit area}$$

Mass per meter of parachute material -> 37 g/m² [\[source\]](#)

Parachute sizing

Primary (Drogue)

Descent rate (>500m): 20 +/- 5 m/s (Typ. 20 m/s)

$$A = \frac{20}{1.22 v^2 C_d}$$

$$A = \frac{16.39}{20^2 C_d}$$

$$A = \frac{0.04098}{C_d}$$

Assuming C_d to be

- 0.8 for a hexagonal chute
- 1.5 for a dome

Hexagonal: $A = \frac{3\sqrt{3}a^2}{2}$

$$A = \frac{0.04098}{0.8}$$

$$A = 0.0512 \text{ m}^2$$

$$a = 14.04 \text{ cm}$$

$$\text{Diameter} = 28 - 29 \text{ cm}$$

Dome: $A = \pi r^2$

$$A = \frac{0.04098}{1.5}$$

$$A = 0.027 \text{ m}^2$$

$$r = 9.3 \text{ cm}$$

$$\text{Diameter} = 20 \text{ cm}$$

Primary Parachute Summary –

	Hexagonal	Dome
Diameter (cm)	29	20
Area (m²)	0.0512	0.027
Mass (g)	1.89	0.999

Secondary (Main)

Descent rate (<500m): 1-3 m/s (Typ. 2 m/s)

$$A = \frac{20}{\rho v^2 C_d}$$

$$A = \frac{16.39}{v^2 C_d}$$

$$A = \frac{4.098}{C_d}$$

Hexagonal: $A = \frac{3\sqrt{3}a^2}{2}$

$$A = \frac{4.098}{0.8}$$

$$A = 5.12 \text{ m}^2$$

$$a = 140.4 \text{ cm}$$

Diameter = 280 – 290 cm

Dome: $A = \pi r^2$

$$A = \frac{4.098}{1.5}$$

$$A = 2.7 \text{ m}^2$$

$$r = 93 \text{ cm}$$

Diameter = ~200 cm

Spill hole-> 5% additional area

Outer Diameter = 210cm

Inner Diameter = 10 cm

Secondary Parachute Summary –

	Hexagonal	Dome
Diameter (cm)	290	200
Area (m²)	5.12	2.7
Mass (g)	189.44	99.9

Shape of parachute

We have picked the **dome parachute** over the hexagonal one due to:

- Lighter weight
- Easier packing
- More consistent deployment
- Availability
- Predictability

CDR refinement

We are planning to do CFD analysis of the entire system before CDR.
This will help us further optimize the parachute's size and shape.

Mechanical Subsystem Design

Mechanical Subsystem Design

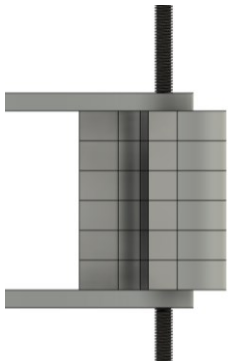
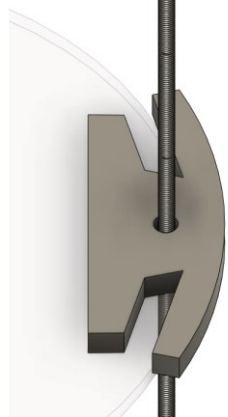
Structural Support Rods:

	1 Rod	3 Rods	6 Rods
Images for Reference			
Reliability	Moderate reliability. Provides basic structural support may not be sufficient to handle significant mechanical stresses, especially under uneven load distribution.	High reliability. The three rods provide a balanced structure, offering good stability and distributing mechanical loads more evenly throughout the CanSat's frame.	Very high reliability. Over-engineered for the weight requirements
Pros	• Lightweight design, contributing to lower overall weight. • Simplified assembly, reducing construction time and potential errors	• Provides better structural support compared to a single rod. • Balanced force distribution, reducing stress on individual components.	• Offers maximum structural stability and resistance to deformation. • Capable of withstanding higher forces, ensuring protection of internal components.

Cons	- Limited structural integrity, making it prone to bending or breaking under stress. - May cause instability during descent due to insufficient support across the body and uneven distribution of mass across body.	- Heavier than the single-rod design, although the weight increase is minimal. - More complex assembly compared to the single rod.	- Heaviest option, increasing mass and potentially reducing mission efficiency. - Excessive redundancy that is not required for the mission scope.
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Selected Design	Rationale
3 Rods	Weight Optimization: The 3-rod design reduces mass compared to 6 rods without compromising structural integrity, making it ideal for the weight constraints of the CanSat.
	Structural Balance: It provides balanced support for even distribution of forces during descent and landing.
	Simplicity: Easier to implement than a 6-rod system, maintaining reliability with a simpler construction process.

Rod and Plate Interfaces:

	Stack Design	Bolt Design
Images for Reference (Side Angle View)		
Images for Reference (Top Angle View)		
Reliability	High reliability due to the integrated mechanical grip and better load distribution. The design's shape helps prevent slippage, providing robust performance in high-vibration environments typical of CanSat deployments.	Moderate Reliability: The right-side design primarily relies on a simple rod and nut fastening system, which is straightforward but may loosen under vibrational loads, especially in high-stress environments like rocket launches or CanSat deployments
Pros	- Better stability and grip on the rod due to the interlocking mechanism. - Distributes the mechanical stress evenly across the surface. - More resistant to rotational forces, reducing the likelihood of misalignment.	- Simpler, fewer parts required for assembly. - Easier to machine or fabricate - Easier to adjust positions and hence is modular
Cons	Requires precise machining to ensure fitment.	- Higher likelihood of loosening over time, especially under vibrational stress. - Less secure, as it relies solely on the tension from the nut.

Selected Design	Rationale
Stack design	Superior Structural Integrity: The stack design's mechanical interlock prevents rotation and displacement of the rod, ensuring stability during high-vibration environments like CanSat launches, making it more reliable than a simple fastener.
	Enhanced Durability: The even distribution of mechanical stress across the structure increases longevity, reducing the risk of failure during critical phases of the mission.
	Justified Complexity: While the design is more complex to manufacture, the added reliability and structural advantages outweigh the minor increase in fabrication difficulty, making it the better choice for mission-critical components.

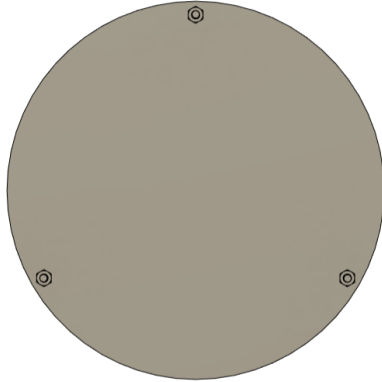
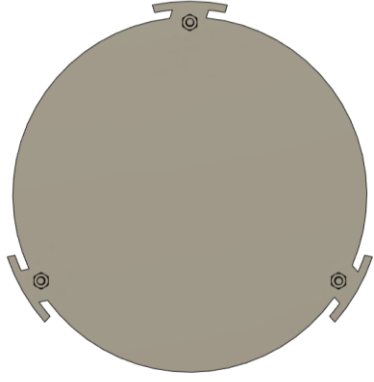
Landing Structures:

	Bendable Legs	Crush Core	Spikes	Landing Legs
Images for reference				
Reliability	Moderate reliability. The legs are designed to absorb impact by bending, which reduces the force transmitted to the CanSat.	High reliability. The crush core is designed to absorb and dissipate energy through controlled deformation, offering predictable and effective impact mitigation.	Moderate to low reliability depending on the landing surface. Spikes may dig into soft ground, effectively absorbing impact, but they could fail on harder surfaces.	High reliability if designed with shock-absorbing mechanisms such as springs or dampers. Good for minimizing impact force across different terrains.
Pros	- Simple design that absorbs impact through deformation. - Lightweight, as legs can be made from simple materials. - Can handle uneven terrain better due to flexibility.	- Efficient energy absorption through compression. - Predictable behaviour during impact. - Can be designed for one-time use.	- Can anchor the CanSat on soft terrain, preventing bounce or roll after landing. - Simple design with minimal moving parts.	- Can incorporate shock absorbers or springs for controlled descent and impact. - Adjustable or flexible designs can handle uneven terrain well.

Cons	• Weakening over time due to repeated stress on the legs. • Limited reusability: once bent, the legs may not return to their original shape.	• Single use: once the core is crushed, it cannot be reused. • Can be more complex to manufacture.	• Inconsistent performance on different terrains. • May cause rebound if the spikes do not penetrate adequately into the landing surface.	• More complex design, potentially heavier than other options. • May require maintenance or more precise manufacturing.
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Selected Design	Rationale
Crush Core	Reliable Impact Absorption: The crush core absorbs and dissipates energy through controlled deformation, offering predictable and highly reliable performance in impact scenarios.
	Consistent Performance: The crush core design provides a uniform response to impacts, ensuring the CanSat's payload is protected in a consistent manner.
	Mission-Critical Focus: Though single-use, the crush core offers excellent performance for one-time deployment, making it the best option for mission-critical landings where reliability outweighs reusability.

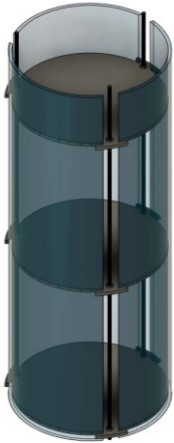
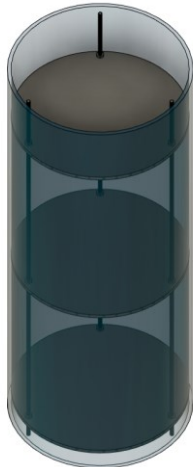
Section Dividers (Plates):

	Without Protrusion	With Protrusion
Images for reference		
Reliability	This design offers a simple, flat interface, which is stable but could be prone to slight misalignments if not secured tightly. Over time, there could be a shift or rotational movement under dynamic loads.	High reliability: The protrusions act as guides that lock the plates into place, preventing rotation or displacement. This feature ensures a more secure fit and better resistance to vibration and dynamic forces.
Pros	- Simple to fabricate and assemble, fewer custom parts are required. - Can be manufactured quickly with less complexity. - Provides a uniform surface for mounting or partitioning internal components.	- The protrusions provide better alignment, ensuring consistent and secure assembly. - Prevents rotational movement and misalignment due to the added interlocking feature. - Offers greater structural stability and strength during high-stress situations like launch or impact.
Cons	Prone to misalignment without additional features like protrusions to guide and lock the plate in place.	Slightly more complex to manufacture due to the need for additional protrusions.

	• Lower resistance to rotational forces or vibrations, which may cause movement over time. • The absence of interlocking features might reduce structural stability.	• May increase material usage slightly, leading to a minor increase in weight or cost.
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Selected Design	Rationale
With Protrusion	Improved Alignment and Stability: The protrusions ensure the plates stay securely in place, preventing rotational shifts or misalignments during vibrations and dynamic forces.
	Better Vibration Resistance: The added interlocking feature improves the plate's ability to withstand vibrations and dynamic loads, making it more reliable in CanSat missions.
	Enhanced Structural Integrity: While slightly more complex to manufacture, the design ensures greater overall structural stability, which is crucial for maintaining internal component organization and protection during high-stress situations like launch and deployment.

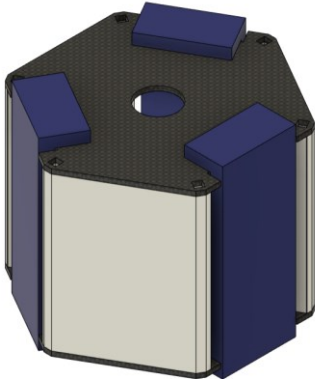
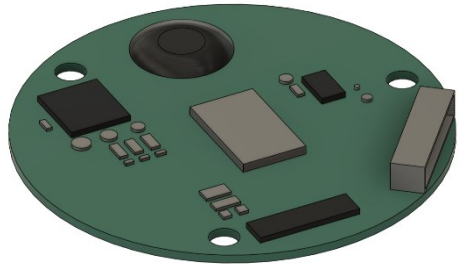
Outer Covering:

	Slider	Wrapper
Images for reference		
Reliability	High reliability: The exposed rods provide direct external structural support, making it easy to access and inspect the frame. This design is rigid and offers excellent resistance to deformation under external stress or during rough landings. The open framework reduces weight and allows for potential modifications or adjustments to the structure.	Moderate to high reliability: Although the external support rods are present, the fully enclosed design provides extra protection for the internal components, shielding them from environmental factors like dust, moisture, or small debris. However, the covering might make it more challenging to inspect or adjust the rods.
Pros	- Easier to access for repairs or modifications, as the exposed rods and open design allow for direct manipulation. - The open structure is lighter, as there's no full outer covering. - External rods still provide structural reinforcement and rigidity.	- Provides additional protection for internal components by fully covering the external rods. - Strong structural integrity with the external rods still providing rigidity, while the outer covering offers a smooth, aerodynamic surface for the CanSat. - Reduces the risk of accidental impact or damage to the exposed rods or internal components.

Cons	• The exposed design could leave the inner components more vulnerable to environmental factors like dust or moisture. • Slightly less protection from external impacts compared to a fully enclosed design.	• Slightly heavier due to the added material required for the full enclosure, which could impact weight optimization for flight • The fully enclosed design makes it harder to access the external rods and internal components for repairs or modifications.
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Selected Design	Rationale
Slider	Weight Optimization: The open design with exposed rods minimizes unnecessary material use, reducing the overall weight of the CanSat, which is critical for flight performance.
	Ease of Maintenance and Modifications: The exposed rods and open structure allow for easier access, making repairs or adjustments simpler during pre-launch checks or after landing.
	Sufficient Structural Integrity: Despite being open, the exposed rods provide enough structural support to maintain stability and protect the CanSat during its mission, making it a balanced choice between weight savings and strength.

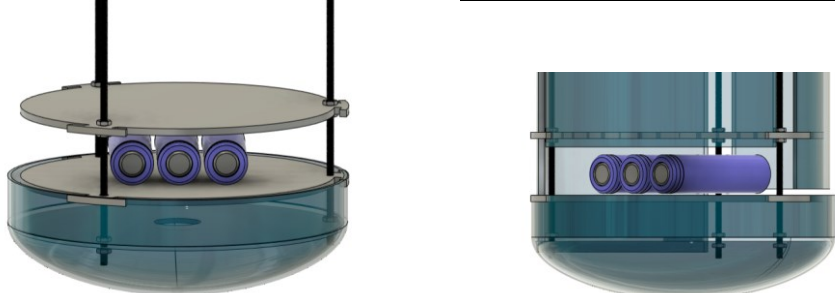
Electronics Placement:

	Compartment	PCB
Images for Reference		
Reliability	Moderately reliable: Though initially easy to implement, the increased reliance on wiring connections introduces more potential failure points, reducing long-term reliability.	Highly reliable: A PCB-based setup minimizes connection issues and improves durability, but it requires careful design to ensure everything is properly integrated.
Pros	• Flexibility: Easier to position and rearrange sensors individually based on space and design needs. • Simpler Prototyping: Requires minimal design and manufacturing effort, making it quicker to implement for testing purposes. • Lower Initial Cost	• Compact Design: Sensors mounted on a single PCB save space, making the design more organized and efficient. • Improved Reliability: Fewer manual connections reduce the chances of wiring faults, enhancing overall system stability and durability. • Reduced Weight: Eliminating excessive wiring lowers the total weight

Cons	• Increased Complexity in Wiring: Numerous wires increase the risk of loose connections, tangling, or signal interference. • Space Management: Wires and separate components can take up more space, leading to cluttered internal layouts and inefficient space usage. • Manual Assembly Risks: Hand-wiring may introduce inconsistencies or errors, leading to unreliable connections and signal issues.	• Higher Initial Cost: Designing and fabricating a custom PCB requires more resources and expertise, increasing costs. • Limited Flexibility and Irreversibility: Once printed, the sensor layout is fixed. Any design errors would require reprinting the PCB, which can be costly and time-consuming.
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Selected Design	Rationale
PCB	• Reliability: The PCB setup is far more reliable, as it minimizes potential connection issues that can arise from loose or tangled wires in the compartment setup. With fewer points of failure, the overall system is more stable.
	• Compactness: The compact design of a PCB allows for efficient use of space, leaving more room for other components or payloads.
	• Weight Considerations: It eliminates excess wiring, reducing the overall weight.

Battery Compartment:

Easily accessible Compartment	
Images for reference	
Description	The CanSat has a slider mesh slider outer covering. This allows quick access to the battery by simply sliding the mesh, eliminating the need for tools. The design ensures efficient maintenance and easy battery replacement when needed.

Materials

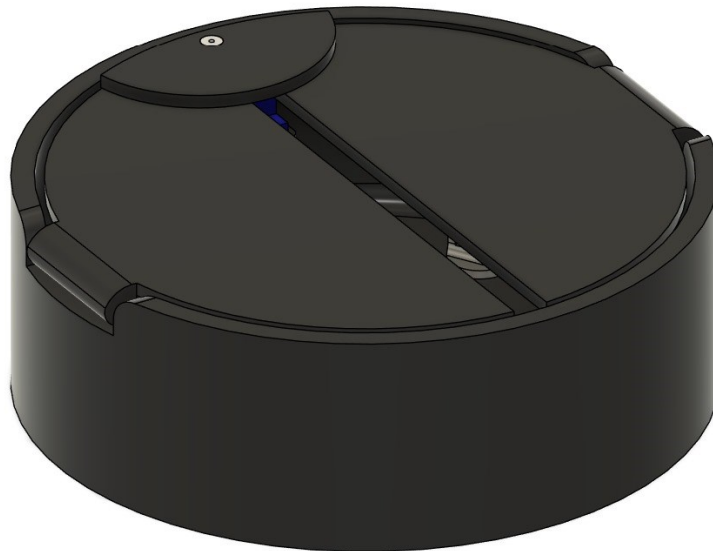
	Material	Pros	Cons	Rationale
Structural Support Rods	Carbon Fiber	Light Weight	Costly, Hard to work with	Rods will be bearing all the load so they must be strong
Structural Support Plates	Acrylic Plexiglass	Easily accessible and Strong	None	It is the easiest material to work with for flat objects that is also strong
Parachute	Nylon Ripstop	Durable and Proven to work	Possibility of Packing issues	Easily available with desired descent rate
Gyroscope Wheels	Steel	Heavy enough	Hard to manufacture	Wheels need to be heavy to be effective
Casings and Containers	PLA Plastic	Durable and Easy to make and test	Increases overall weight	Easily available and relatively cheap

Deployment Mechanism Design

Deployment Mechanism Design

From the Descent Control Analysis, we know that we have to use two parachutes for the primary and secondary descent control systems.

This rough design outlines the structure and functionality of the deployment mechanism, which will be positioned at the top of the CanSat.



Primary Parachute

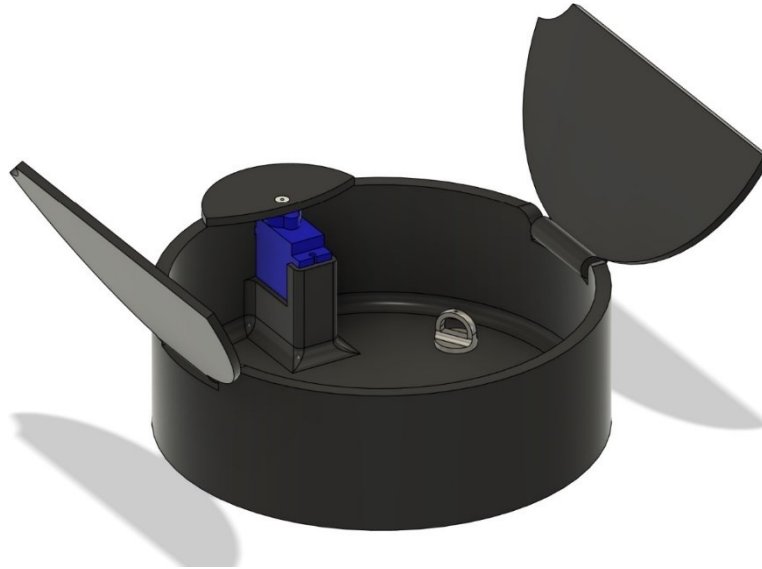
The primary parachute is responsible for the initial phase of descent after the CanSat is released from its deployment vehicle. It is designed to deploy automatically without the need for any active systems. As soon as the CanSat is deployed, the small primary parachute, which is packed loosely, unfurls due to air resistance and begins to slow the descent.

This passive deployment reduces the risk of failure that comes with added complexity.

Secondary Parachute

Unlike the primary parachute, the second parachute is deployed using an active system so that we have control over the time of release. Two spring loaded doors hold it in place. At 500m, the doors are opened, and the primary parachute pulls out the secondary parachute, which is packed in such a way that it unfurls completely and efficiently.

The secondary parachute is packed in such a way that the spill hole is at the top of the release mechanism. The spill hole is in turn attached to the primary parachute for quick deployment.



The three rods that run through the whole body of the CanSat are used as the attachment point to this mechanism. A 3d printed variant of wing nuts that have a hole in them is used to connect the secondary parachute wires to the CanSat firmly. This is a very basic design that will be refined after testing.

Other Mechanisms

Some other designs that we considered are:

1. One door – There could be issues with tangling with the parachute wires. Also, in this design the parachute would leave the CanSat in an uneven fashion and might not unfurl properly.
2. Two compartments – In this design we would have two compartments, and the primary and secondary parachutes would be in each one. This option is an added level of complexity that is not needed. It might also cause imbalance during descent.

Mechanical Stabilisation System

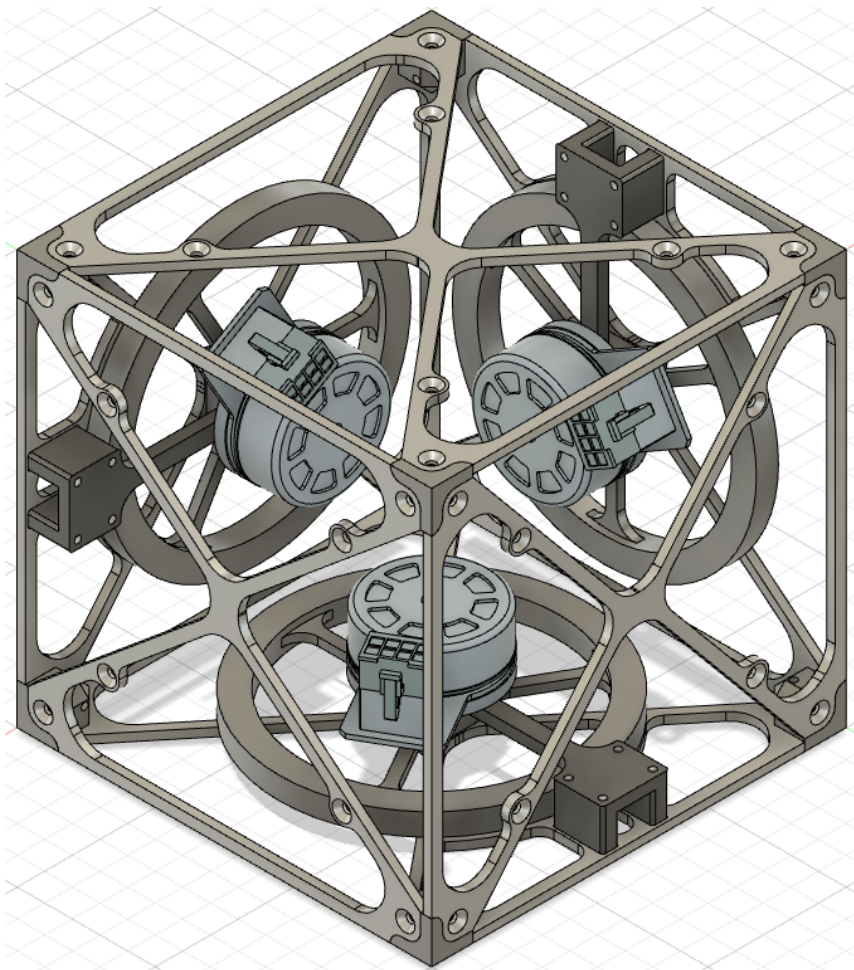
Mechanical Gyro Stabilisation System

Abstract

We are creating a reaction wheel system to stabilise roll, pitch and yaw of our rocket. When a torque is applied on the CANSAT due to any sort of disturbance, an opposing torque is applied by spinning wheels to correct the orientation of the CANSAT and maintain the roll as well.

Design Overview

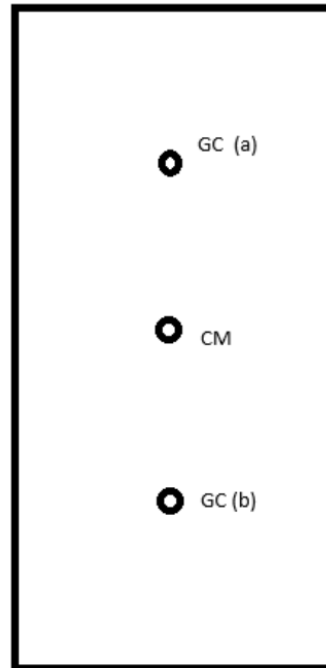
The gyro module will be made as modular as possible, so that it can be made independently of other components. It will be a simple box type structure that will fit into the CANSAT, with motors on 3 faces attached using supports. The gyro motors will use a PID control algorithm



Abbreviations: GC = Gyro Centre, CM = Centre of Mass, MI = Moment of inertia

Design Decisions

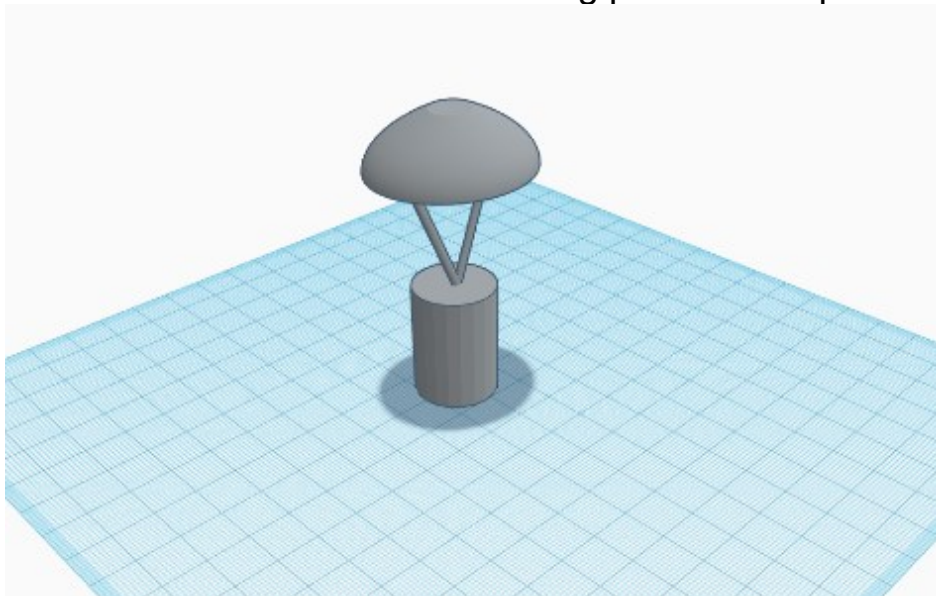
a) Placement of gyro module and Centre of mass position



Because of the effect of the wind and the gyro centre and there are two cases that can arise:

1. The Centre of Mass of the Gyro is above or below the centre of mass of the CanSat body, then the torque due to the wind is higher than normal.
2. If the Centre of Mass of the Gyro is at the Centre of Mass of the Body (Centre of Mass at the centre of the body), the torque due to the wind is significantly reduced.

We also have the effects due to the bottom landing part and the parachute.

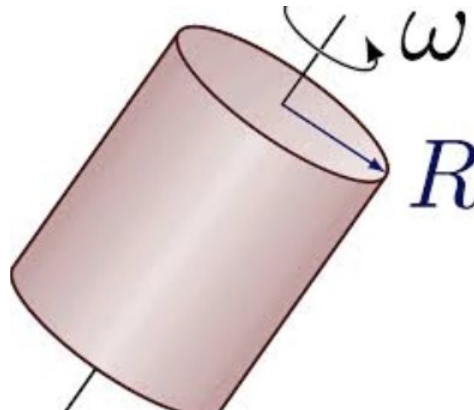


If C.M is high, then the bottom will be very unstable due to the wind.

Therefore, to lower the movement at the bottom and reduce the effect of wind torque at the same time, we aim to have a lower CM of the CanSat approximately 7cm below the centre of the CanSat.

But Not Too low, because that would also make the top wobbly and hence increase work for the gyro.

Requirement Calculation



a) Inertia through the cylinder for roll

$$I_{through} = I_{outerBody} + I_{others}$$

Let us consider a reduced mass of $M = 800\text{gm} \rightarrow M' = 400\text{gm}$ due (for mass of components) due to uniform spread in disc like behaviour and let the body mass be 200gm .

Therefore

$$I_{through} = (200 + 400)r^2$$

$$I_{through} = 0.00294 \text{ kg} - \text{m}^2$$

b) Inertia perpendicular to the cylinder for pitch, yaw

We are considering the effect due to the following items:

- I. Battery: 180gm , $r = 4\text{cm}$
- II. Gyro: 220gm , $r = 10\text{cm}$
- III. Electronics: 200gm , $r = 12\text{cm}$
- IV. Parachute system: 70gm , $r = 15.5\text{cm}$
- V. Lander system: 70gm , $r = 17.5\text{cm}$

Inertia due to all components = $0.00919 \text{ kg} - \text{m}^2$

Inertia due to the body (assuming cylinder): $(MR^2/12) = 0.0015 \text{ kg} - \text{m}^2$

Total perpendicular inertia about the centre = $0.01 \text{ kg} - \text{m}^2$

Perpendicular moment of inertia about gyro centre = $0.01 + 1 * (0.05) = 0.013 \text{ kg} - \text{m}^2$

Importance of roll axis stabilisation

If roll axis is not stabilised, then what will happen is that the state of the pitch and yaw axis will have to be exchanged and reversed continuously as the body spins. Hence Limiting roll is extremely crucial to the proper function of pitch and yaw stabilisation.

Calculations to find required motor torque

Note: saturation limit here would imply max RPM.

There are a lot of factors that can try to change the orientation of the CANSAT.

- a) CANSAT is not balanced properly. (i.e. it tilts slightly when hung normally by a thread where parachute is connected)
- b) Wind turbulence
- c) If fins are used for stabilisation, then any error in fin placement will also induce tilt and roll.

Now let us start with pitch and yaw correction.

Let us consider a 50g-force equivalent acting 15 cm from the GC for 1 second and let us consider that it is exactly along one gyro only for simplicity's sake, that gyro wheel responds in 't' time to counter the torque.

Amount of tilt induced will be more depending on the time t.

$$\theta = \frac{1}{2} I \alpha t^2$$

$$\tau = F * d = 0.0735 \text{ N} * \text{m} \quad \alpha = \frac{\tau}{I} \quad \alpha = 5.6 \frac{\text{rad}}{\text{s}^2} \quad \theta = 0.0364 t^2 \text{ rad} = 2 t^2 \text{ deg}$$

Reaction time 't' depends on the entire systems response, There will be a delay caused by the processor and the motor driver, and then the angular velocity accumulated during this time will have to be removed as well by the motor which is lesser if the motor has more torque.

This is only for a sample force, actual force or lower, and it will be much more continuous as well.

This gives us an idea to find a motor. We need to have a motor with a torque in the order of magnitude of the mentioned torque and try to find motors with higher max rpm (also bidirectional).

For the roll motor, if we are not using fins then roll induced will be much lesser and hence, we should get a motor with a much higher saturation while sacrificing torque. However, if using fins, then roll induced will be much more due to the lower moment of inertia as well, hence we will need something that reacts immediately.

Control Algorithm and desaturation strategy

For pitch and yaw, we will divide the tilt angle into components for the pitch and yaw gyroscopes. We will use a simple PID algorithm to control the gyro.

When the tilt is very less, say within 3 degrees, the speed of the pitch, yaw motors can be bled very slowly. This works because the cansat has some inertia and even external forces from the airflow around the body, tugging of the parachute (and the fins also if implemented) that usually help the cansat to retain its orientation.

Similarly, the speed of roll gyro should also be bled when there is no rotational angular velocity, also provided that the speed of roll, pitch gyros is also very less.

In the case of very high saturation, slightly increase the bleed threshold so that motors get desaturated faster.

Operation strategy:

Start the gyros a few seconds after CANSAT is ejected from the rocket.

A few seconds after the secondary descent mechanism is activated increase the bleed threshold to 15 degrees from the normal 3 degrees and then also increase bleed rate, so that we can bleed out all the speed accumulated that will inevitably be accumulated due to the shock of releasing the secondary descent machine (parachute). When the gyros are desaturated, change the threshold and bleed rate back to normal again.

Gyro wheel weight and design:

Ring/wheel provides the highest Moment of Inertia per unit mass

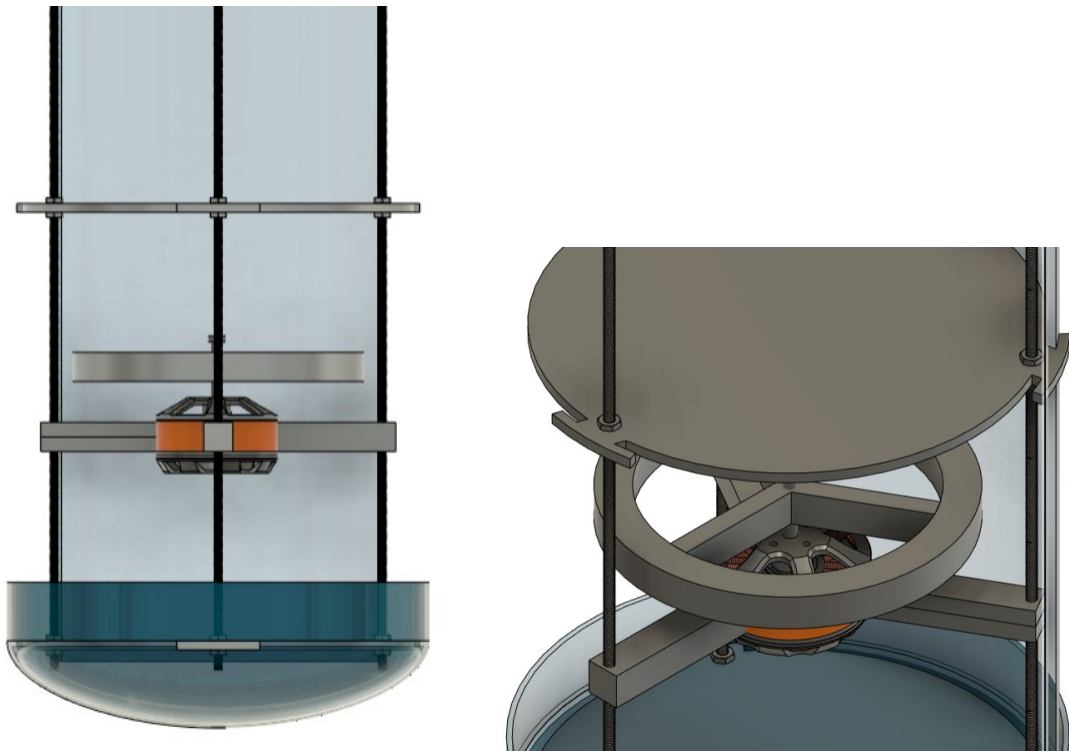
$$I = MR^2$$

Therefore, we should create the wheel structure such that the radius as as high as possible and most of the mass is concentrated at the furthest possible distance from the centre.

Why do we need to maximise I ?, because more moment inertia means that the gyro systems saturation limit is increased. $Saturation \propto I$

What should be the weight of the wheels? It should be the maximum weight possible, all the extra weight available in our budget should be used up by the wheels eventually. Also, if the weight of the wheels is very less than the gyro motor MAX RPM should be given much more priority than its torque.

Till this point we have only talked about a reaction wheel design which is our main consideration for now, but given below is an alternate design based on a completely different principle.



Mechanical gyro that only uses gyroscopic effect

This is a backup design which uses the gyroscopic effect to stabilise the rocket, ofcourse this comes at the cost of lot of roll, but roll is acceptable here since there is no pitch, yaw reaction wheel stabilizers.

This has a major flaw that is why we mostly are not considering this idea.

If we start it from the launch pad, then it tries to retain whatever state it gets when it gets ejected, however that state will still come to a stable equilibrium somewhere around the right orientation and it will still be somewhat successful at combating small tilts and vibrations.

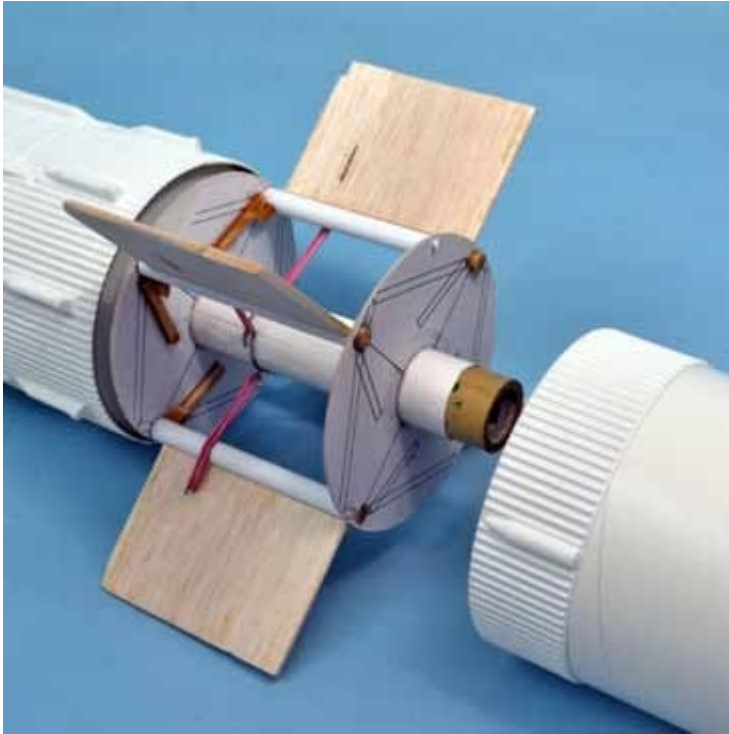
If we start it after ejection then there is a lot of roll which is in the opposite direction creating its own effect which cancels out the effect due to the gyro and hence there is no overall stabiliation effect, unless the outer speed is bled off or contained by connecting the body directly to the parachute with a large ring in between the cansat and the parachute to prevent tangling.

Both these ideas also require full speed continuous functioning of the motor which is very inefficient as well.

Fins for stabilisation

Fins are a very effective, tried and tested method for stabilising the CANSAT during descent

We can make the fins such that they flip out as soon as the CANSAT is ejected



[Source](#)

When the CANSAT tilts, more air hits the fins and hence the fins exert an opposite force trying to correct the tilt.

However compared to the gyro this is more of a rough sort of solution and is much more useful during the initial 1km – 500m descent phase due to its higher effectivity in higher airspeed.

The worst downside being that any imperfections will induce roll continuously which might be detrimental to pitch, yaw gyros, imperfections will also slightly affect pitch and yaw.

Fins are like a rough tool, whereas the gyro is a much more fine tool. This will also reduce the efficacy of the gyro. However it is much more reliable, especially so for initial descent.

Given the heavy weight and low air speed, we should have 7-10 cm long fins fully wrapped. (Note this does affect our standard design quite a lot, increases complexity)

This a good idea but we will only implement it if we have enough space, mass left.

Power Budget

Power Budget

Subsystem	Component	Voltage (V)	Current	Power Consumption	Duty Cycle (in 1 hour)	Total Power
Sensor Subsystem	LC89 HA	3.3	99 mA	326.7 mWh	2	653.4 mWh
	MPU 6050	3.3	4.1 mA	13.53 mWh	2	27.06 mWh
	BMP 280	3.3	720 uA	1.296 mWh	2	2.592 mWh
	BME 680	3.3	3.7 uA	12.21 uWh	2	24.42 uWh
	HDC 1080	3.3	710 nA	2.343 uWh	2	4.686 uWh
	INA 260	3.3	310 uA	102.3 mWh	2	204.6 mWh
	ADXL377	3.3	300 uA	99 mWh	2	198 mWh
Gyroscope	Motors*3	11.1	300*3 = 0.9 A	9.99 Wh	2	19.98 Wh
Communications	SD Card Module	5	32 uA	10.56 mWh	2	2.11 mWh
Payload	PAS OUT 01	5	70 mA	350 mWh	2	700 mWh
	Camera	5	650 mA	3.25 Wh	2	6.5 Wh
	DS3231	3.3	200 uA	66 mWh	2	132 mWh
	XBEE 3 PRO	3.3	135 mA	445 mWh	2	890 mWh
Controller	ESP32-WROOM-32E	3.3	240 mA	792 mWh	2	1.584 Wh
Total						30.87 Wh

Mass Budget

Mass Budget

(*Supplier – Robu / Local Vendor / Amazon / Mouser / etc.)

Components		Source	Unit Mass (g)	Quantity	Total Mass (g)
Structural Subsystem	Threaded rods	Supplier	30	3	90
	Brackets for holding	Estimates	15	2	30
	Plates	Estimates	15	4	60
	Subsystem Total				180
Parachute Deployment System	Container	Estimates	15	1	15
	Parachutes	Supplier	10+90	1+1	100
	Attachment points	Supplier	10	1	10
	Servo motor	Supplier	11	1	11
	Subsystem Total				136
Stability Control Mechanism	Gyro Wheels	Calculations	70	3	210
	Motor	Supplier	50	3	150
	Mountings	Calculations	20	3	60
	Subsystem Total				420
Landing Legs	Legs / Impact absorbers	Calculations	20	1	20
	Subsystem Total				20
Outer shell	Outer mesh covering	Calculations	5	3	15
	Subsystem Total				15

CDH Subsystem	XBEE Pro S2C	Supplier	7	1	7
	ESP-32	Supplier	10	1	10
	Antenna	Supplier	5	1	5
	Subsystem Total				22
Electrical Subsystem	Battery Pack	Supplier	50	3	150
	Buck Converter	Supplier	10	1	10
	Mounts and Wires	Calculations	15	--	15
	Subsystem Total				175
Sensor Subsystem	LC89 HA	Supplier	8	1	8
	MPU 6050	Supplier	5	1	5
	BMP 280	Supplier	6	1	6
	BME 680	Supplier	6	1	6
	HDC 1080	Supplier	4	1	4
	INA 260	Supplier	5	1	5
	ADXL 377	Supplier	4	1	4
	Ada Fruit Spy Cam	Supplier	5	1	5
	Air Quality	Supplier	20	1	20
	Subsystem Total				63
Miscellaneous	Margin for errors	-	-	-	10
Total					1041

Mass Budget Summary –

Sr. no.	Subsystem / Components	Mass (g)
1.	Structure	180
2.	Parachutes	136
3.	Stability control mechanism	420
4.	Landing legs	20
5.	Outer covering	15
6.	Sensors	63
7.	Communications	22
8.	Electrical power	175
9.	Miscellaneous (Buffer/Margins)	10
Total		1041

Hence, we find that the total mass is well within the margins specified by the mission guide. Nonetheless we have the following measures planned to avoid going over/under mass –

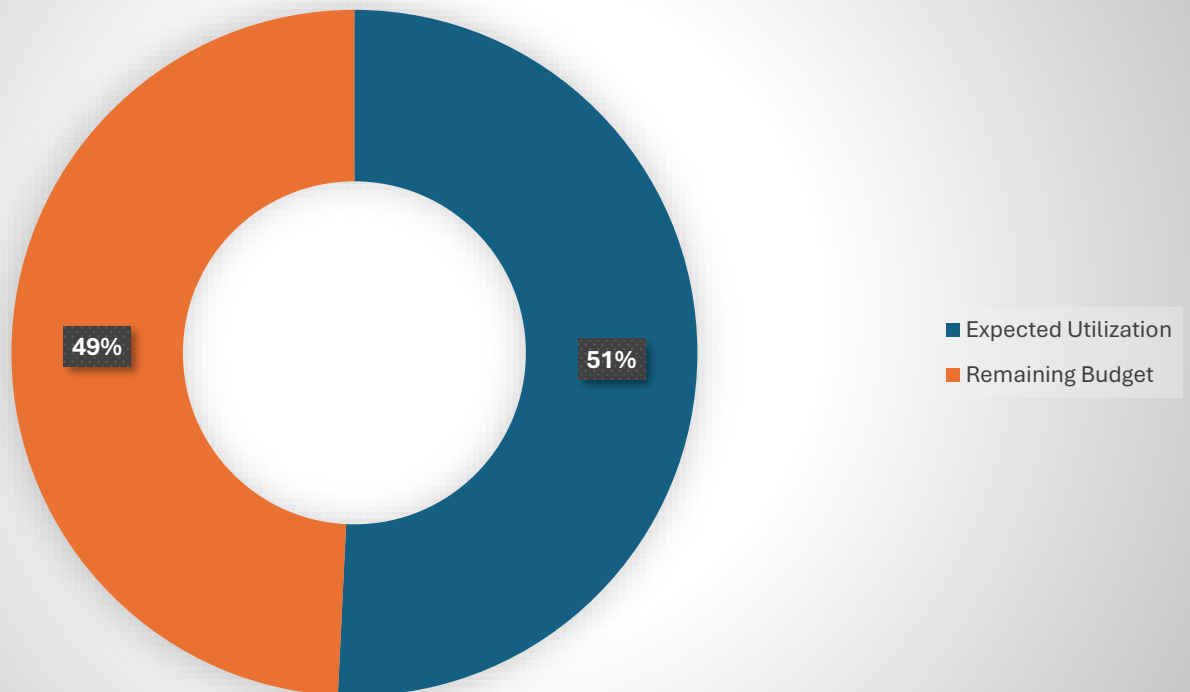
- To lower the infill of 3-D printed parts to reduce mass
- To have holes in the plates and containers.
- To add the mass by adding vestigial screws and bolts.

Hardware Budget

Hardware Budget

Sr. no.	Components	Reuse (Yes/No)	Unit Price (INR)	Quantity	Total Price (INR)
1.	3D printed parts	No	5/g	300g	1500
2.	Structural Rods	Yes	300	3	900
3.	Acrylic Plates	Yes	20	4	80
4.	Servo	Yes	100	1	100
5.	SD Card	Yes	369	1	369
6.	XBEE Pro S2C	Yes	1999	2	3998
7.	Battery Pack	Yes	100	3	300
8.	Switch	Yes	20	1	20
9.	LEDs	Yes	20	5	100
10.	Buzzers	Yes	100	1	100
11.	Custom PCB	No	7000	1	7000
12.	Parachutes	Yes	2000	2	4000
13.	Ball Bearing	Yes	50	1	50
14.	Screws and Threads	Yes	200	-	200
15.	Mylar Streamer	No	50	1	50
16.	Sensors	Yes	30000	-	30000
17.	ESP-32	Yes	1000	1	1000
18.	Wires, Mounts, Etc	Yes	500	-	500
19.	Screws, Adhesive, etc	Yes	500	-	500
Total					50767

Hardware Budget



From the total 1,00,000rs we could use we are expected to utilize 50,767rs, which is roughly 51% and have a buffer of 49,233rs. We will start looking for external sponsors in the upcoming days and we might get a sponsorship for ease in materials procurement. Other costs will include travelling to the launch site and possible shipping to the launch location which currently team members will sponsor themselves.

Integration and Testing

Integration and Testing

Integration and Testing Stages	
Subsystem Level Test Plan	Test individual components/subsystems e.g. Sensors, CDH, EPS, GCS etc. to identify and troubleshoot before integrating into a larger system.
Integrated Level Test Plan	Testing for Integrated Systems e.g. Descent Control, Electro-mechanical systems, Communications to identify and troubleshoot any issues.
Environment Test Plan	With a fully integrated CanSat, we test the entire assembly according to environmental testing standards set by the mission guide.
Simulation Test Plan	The CanSat is commanded in Simulation mode and the simulated launch data is sent to Ground station to demonstrate FSW, CDH.

Integrated Level Functional Test Plan	
Descent Testing	• Ensure parachute descent speeds are accurate to 1 m/s of desired speeds • The secondary descent control mechanism starts operating at the desired height
Communications	• Test long distance radio communication between XBEEs. • Test two-way communication Test for optimal antenna position. • Test Ground Control Station's GUI display
Mechanisms	• Test simulation mode software with given competition datasets. • Ensure simulation mode commands are transmitted through two-way communication. • Ensure simulation mode data is received by the ground station.
Deployment	• Test the parachutes to assure they won't become tangled on release. • Test tolerances between parachutes, CanSat, and rocket. • Test the tensile strength of parachute chords via environmental drop tests.
Simulation	• Test simulation mode software with given competition datasets. • Ensure simulation mode commands are transmitted through two-way communication. • Ensure simulation mode data is received by the ground station

Environmental Testing

Environmental Test Plan	
Drop Test	- The goal of this test is to ensure that the CanSat can survive 30Gs of shock - We will attach the parachute and drop it to simulate the shock of the CanSat leaving the rocket body.
Thermal Test	- The goal of the thermal test is to ensure that the entire CanSat can operate successfully in a 60 °C environment. - We will build a thermal chamber to contain the CanSat and place the functioning CanSat into the 60 °C chamber for a duration of two hours
Vibration Test	- The vibration test aims to verify the mounting integrity of all components, mounting connections, structural integrity and battery connections. - The CanSat will be attached to an orbital sander which will be cycled for five intervals of five seconds.
Fit Check	- The fit check is to ensure that the CanSat will fit in the rocket on flight day and deploy without issues. - Construct a rocket payload section and test tolerances between the payload wall and CanSat
Vacuum Test	We will construct a vacuum chamber and then simulate the air pressure changing to ensure that all systems and flight software is working

Mission Operation and Analysis

Mission Operation and Analysis

Launch Day Sequence of Events-

Event	Tasks	Teams involved
Arrival	Arrival at the launch site with CanSat and Ground Station Hardware	ALL
Pre-Launch Events	Assemble CanSat and Mechanical System Integration	Cansat Crew
	Ground Station Setup and Antenna Calibration	Ground Control
	Calibrate and Test Electrical Systems	Cansat Crew
	Turn CanSat on	Cansat Crew
	CanSat Integration into Rocket	Cansat Crew
Launch Events	Move to Launch Control Table	ALL
	Initiate CanSat Launch	Mission Control Officer
Descent	Monitor the Data Reception from Telemetry and Validate Flight State Changes	Ground Control
Recovery Phase	Locate and Recover the CanSat using GPS and Audio-Visual Trackers	Recovery Crew
Submission Phase	Submit the CanSat, SD Cards and Flight Data to the Field Judge	Ground Control & Recovery Crew
Data-Analysis	Analyze the Flight and Telemetry Data	Ground Control
	Post Flight Review (PFR)	
	Return to Check-in	

In addition to that we will have a sticker on the CANSAT Body which will display the contact information in Hindi, English and the regional language of the Launch Site. We will also have several mission rehearsals before launch and have a Mission Operations Manual prepared for everyone in the team so that they can perform their duties efficiently.

Location and Recovery Plan

ocation and Recovery Plan

We plan to investigate multiple ways in which we can locate the CanSat:

1. Audible buzzer (>92 dB)
2. Light
3. Radio beacon
4. WiFi based
5. Bright Florescent colours

Audible buzzer (>92 dB)

We found this buzzer that satisfies the requirements.

RS PRO 92dB Panel Mount Continuous Internal Buzzer

Cost: INR 700/-

Voltage: 3-20 V

Sound Level: 92 dB

Frequency is dependent of the voltage.

Datasheet: [Link](#)



We can vary voltage to create a pulsing tone that is more distinct and less likely to blend in.

Light

4 pulsing LED for early morning recovery.

Maybe some wavelength that is not present in environment naturally.

GSM/GPRS

When the XBEE signal is probably lost due to no line of sight, this can still work, because this does not require full line of sight and also has a higher peak transmission power. This enables the transmission of the location data and all other packets. It can also provide location as a backup to the GPS system, which can also fail if blocked.

Radio beacon

<https://www.nar.org/fai-spacemodeling/construction-techniques/recovery-beacons/>

We are also planning to integrate a simple radio beacon that send a coded frequency pulse every few seconds.

This can be used to find the location.

WiFi based

We can use the ESP32 onboard WiFi to broadcast on a channel and then use a WIFI detector app on a phone to find the location of the CanSat.

This will work well if the surroundings are non-cluttered

Bright colour (of the tricolour)

We will print the 3d print parts out of coloured filament and add a reflective clear coat. If we are not able to do that (due to structural constraints), we will add a coat of abrasion resistant clear paint to boost visibility. [\[Link\]](#)

Program Schedule

Program Schedule

Name	Legend
Harsh Kapoor	HKR
Madhan Sai Krishna	MSK
Anushka Sharma	ANS
Adithya Kishor	ADK
Aditya R Shankar	ARS
Tejasvini Ramaswamy	TRS
Vignesh V Manikantan	VVM
Vijay Aravynthan S R	VAS
Mechanical Team	MHT
Electrical and Sensor Team	EST
Communications Team	CMT
Management Team	MGT
All members (Team Brahmand)	ALL

Management Team	Team Members	Start Date	End Date	No. of Days
Team Member Recruitment	MGT	1/6/24	20/6/24	20
First Meeting of the Team	ALL	25/6/24	25/6/24	1
Analysis of the Mission Guide	ALL	25/6/24	30/6/24	5
Planning and Work distribution	ALL	30/6/24	15/7/24	15
Scouting for resources and permissions	MGT	25/6/24	10/7/24	15
Sub – team Division	ALL	15/7/24	30/7/24	15
Quiz – 1 Examinations	ALL	25/8/24	29/8/24	5
Mid – Sem Examinations	ALL	19/9/24	25/9/24	7
PDR Preparations and Submissions	ALL	25/9/24	30/9/24	5

Mechanical Team	Team Members	Start Date	End Date	No. of Days
Analysis of Mission Guide	MHT	1/7/24	4/7/24	4
Analysis of CONOPS	MHT	5/7/24	9/4/24	4
Payload placement and planning	MHT	10/7/24	14/7/24	4
High level design of Payload	MHT	15/7/24	20/7/24	5
Structure placement and planning	MHT	21/7/24	24/7/24	4
High level design of structure	MHT	25/7/24	30/7/24	5
Trade-offs for the body	MHT	31/7/24	3/8/24	4
Trade-offs for the Parachute	MHT	4/8/24	8/8/24	4
Descent Rate Initial Estimates	MHT	9/8/24	13/7/24	4
Secondary Descent Control Design	MHT	14/8/24	18/8/24	4
Mechanisms placement and planning	MHT	19/8/24	22/8/24	3
High level design of mechanisms	MHT	23/8/24	28/8/24	5
Landing legs placement and planning	MHT	29/8/24	31/8/24	3
High level design of landing legs	MHT	1/9/24	5/9/24	4
Electronics placement and planning	MHT+EST+CMT	7/9/24	9/9/24	2
High level design of electronics box	MHT+EST+CMT	10/9/24	14/9/24	4
Integration of designs	MHT+EST+CMT	15/9/24	20/9/24	5
Final checks and discussions	ALL	21/9/24	24/9/24	3

Electrical and Sensor Team	Team Members	Start Date	End Date	No. of Days
Analysis of Mission Guide	EST	7/7/24	10/7/24	3
Analysis of CONOPS	EST	11/7/24	13/7/24	2
Discussion on Mission Requirements	EST	14/7/24	16/7/24	2
Distribution of work among team	EST	17/7/24	18/7/24	1
Searching Sensors and Modules	EST	19/7/24	31/7/24	11
Discussions on Trade-offs	EST	1/8/24	3/8/24	2
Discussions on bonus objectives	EST	4/8/24	8/8/24	4
Scouting for Sensors in College	EST	9/8/24	14/8/24	5
Discussions on RIS as payload	EST+CMT+MGT	15/8/24	22/8/24	7
Finalizing a preliminary list	EST	23/8/24	28/8/24	5
Discussions for placement	EST+MHT+CMT	7/9/24	20/9/24	11
Final Checks and discussions	ALL	21/9/24	24/9/24	3

Communications Team	Team Members	Start Date	End Date	No. of Days
Analysis of Mission Guide	CMT	1/7/24	5/7/24	5
Analysis of CONOPS	CMT	5/7/24	9/4/24	4
Discussion on Mission Requirements	CMT	10/7/24	14/7/24	4
Distribution of work among team	CMT	15/7/24	20/7/24	5
Discussions on Central Computer	CMT	21/7/24	31/7/24	10
Discussions on Radio Module	CMT	25/7/24	5/8/24	10
Discussions on Data Storage	CMT	10/8/24	25/8/24	15
Discussions on Flight Software	CMT	10/8/24	30/8/24	20
Discussions on Ground Control	CMT	15/8/24	5/9/24	20
Working on Flight Algorithm	CMT	10/8/24	25/8/24	15
Selection of Components and Trade-offs	CMT	1/9/24	6/9/24	5
Discussions for placement	EST+MHT+CMT	7/9/24	20/9/24	11
Final Checks and discussions	ALL	21/9/24	24/9/24	3

Conclusion

Conclusion

Major Accomplishments:

Management Tasks:

- Secured permission to utilize labs for building and testing the CanSat models.
- Finalized most of the required components, which are now ready for procurement.

Mechanical Team:

- Developed multiple conceptual designs.
- Completed basic CAD modeling for all major components.

Electrical and Sensor Team:

- Nearly finalized the selection of sensors and major electrical components.
- Received approval to borrow essential sensors from the college's inventory.

Communications Team:

- Successfully procured the necessary XBEE modules.

What's Next:

Team Brahmand has explored various innovative ideas and considered numerous trade-offs for each component of the CanSat. While not all ideas can be listed here, we plan to move forward with rigorous testing to evaluate and refine our designs.

Once key ideas have been tested and trade-offs better understood, we will prepare the PPT for the Preliminary Design Review (PDR) presentation, highlighting the major decisions and trade-offs identified during this phase.

Appendix

A. GPS

1. Difference between GPS starts

TFFF – Time to First Fix is the process of time in which it takes a [GPS](#) enabled device to obtain signal from enough [satellites](#) to determine accurately its location.

Data	Cold start	Warm start	Hot start
Position	-	<100km	accurate
Visibility of any GPS satellites	-	Almanac is valid	accurate
Velocity	-	<25 m/s	accurate
Time	-	<20 secs	accurate

Almanac = ephemeris data (valid for 4 hrs, broadcasted every 30 mins)

Hence the time that we need is the **hot start time**.

External source: [gpsbeam](#)

2. GPS accuracy

Measurement scales

Term	Definition	Precision
CEP	Circular Error Probable	50%
RMS	Root Mean Square	63-68%
2DRMS	Two times the distance of RMS	95-98%
R95	Radius 95%	95%

Vertical Accuracy

Most receiver manufacturers do not provide a vertical (altitude/elevation or z-value) accuracy specification. Just as a general observation, the vertical accuracy of GNSS/GPS receivers is typically 1.7 times the horizontal accuracy. For example, a receiver with 1 m 2DRMS horizontal accuracy would likely provide closer to 2m vertical

accuracy. Again, this estimate is based on general observation of several different receivers, not thorough testing of any one receiver.

External source: [Juniper systems](#)

B. Bibliography

Datasheets

GPS

- [Neo M9](#)
- [LC89 HA](#)
- [ELNMC1A](#)
- [BD9250s](#)
- [Venus827](#)

IMU

- [BNO055](#)
- [MPU-6050](#)
- [MPU-9250](#)
- [LSM6DS3](#)
- [LSM6DS0](#)

Barometer

- [LPS22HB](#)
- [BMP388](#)
- [BMP390](#)

Temperature

- [MCP9808](#)
- [DHT22](#)
- [BME680](#)

Current sensor

- [INA260](#)

Accelerometer

- [ADXL377](#)

Particle sensor

- [Honeywell impact 2.5 PM](#)
- [SN-GCJA5](#)
- **PAS OUT 01** [[Datasheet](#)]

- SDS011 [[Datasheet](#)]
- MQ135 [[Datasheet](#)]
- **MQ131** [[Datasheet](#)]

Reconfigurable Intelligent Surface (RIS)

[[etsi.org](https://www.etsi.org)]

MicroSD

https://documents.westerndigital.com/content/dam/doc-library/en_us/assets/public/sandisk/product/memory-cards/ultra-uhs-i-microsd/datasheet-ultra-uhs-i-microsd.pdf

Antenna

<https://www.3antelecom.com/2-4ghz-wifi-wlan-12dbi-antenna-sma-male-connector-6053889.html>

https://media.digikey.com/pdf/Data%20Sheets/Adafruit%20PDFs/4622_Web.pdf

<https://www.digi.com/resources/documentation/digidocs/pdfs/90000976.pdf>

GSM

[SIM800L](#)

Radio beacon

<https://www.nar.org/fai-spacemodeling/construction-techniques/recovery-beacons/>

Acronyms

Scientific Acronyms

Abbr.	Full Form
A	Ampere
°C	Celsius
dBi	decibel
g	Accel. due to gravity
G	G-force
GHz	Gigahertz
h	hour
hPa	Hecta Pascal
Hz	Hertz
km	kilometer
kPa	Kilo Pascal
KB	Kilo Byte

Abbr.	Full Form
m	meter
mA	Milli-Ampere
mm	milli meter
mW	milli Watt
MB	Mega Byte
MHz	Mega-Hertz
μA	Micro-Ampere
N	Newton
Pa	Pascal
s	second
V	Volt
Kg	Kilo gram

General Acronyms

Abbr.	Full Form
A	Analysis
CAD	Computer-Aided Design
CAN	Controller Area Network
CDH	Communication & Data Handling
CDR	Complete Design Review
CIT	CanSat Integration & Testing
CoG	Centre of Gravity
CoM	Centre of Mass
CONOPS	Concept of Operations
D	Demonstration
DCS	Descent Control Strategy
EPS	Electrical Power Subsystem
GCS	Ground Control Station
GPS	Global positioning system
I	Inspection

Abbr.	Full Form
IC	Integrated Chip
I ² C	Inter Integrated Circuit
IDE	Integrated Development Environment
INR	Indian National Rupee
LN	Low Noise
LP	Low Power
MAAC	Motor Actuated Angle Control
MCU	Microcontroller Unit
OS	Operating Systems
PAC	Passive Angular Control
PDR	Preliminary Design Review
T	Testing
UHS	Ultra-High Speed
ULN	Ultra Low Noise
UTC	Universal Time Coordinated

THANK YOU
